

DATA CENTER PROJECT

Master Technical, Business Case, Project Management, and Operations
Reference

Prepared for the Engineering and Technical Delivery Team

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1. Executive Summary

This document presents the complete technical design, business case, project management plan, and operations and maintenance manual for the Data Center Project ("the Project"). It is intended as the master reference for the engineering and technical teams responsible for designing, building, commissioning, and operating the facility through its full lifecycle.

The Project delivers a Tier III-equivalent, concurrently maintainable data center facility sized to support enterprise production workloads, with a phased capacity model that allows compute, storage, and network infrastructure to scale from an initial 1 MW critical IT load to a build-out capacity of 4 MW across four phases. The design targets an annualized Power Usage Effectiveness (PUE) of 1.35 or better, N+1 redundancy at the facility level with 2N redundancy on critical UPS and cooling paths, and a Recovery Time Objective (RTO) of under four hours for core services.

The document is organized into six parts: (I) Project Overview and Business Case, (II) Facility Design and Infrastructure, (III) IT and Network Infrastructure, (IV) Project Management Plan, (V) Operations and Maintenance, and (VI) Appendices. Part III, covering server, storage, network, and cabling architecture, is presented in the greatest depth, reflecting the primary audience of this document: the engineering and technical delivery team.

1.1 Key Objectives

- Deliver a highly available, concurrently maintainable data center meeting Uptime Institute Tier III criteria and TIA-942 Rated-3 structured cabling standards.
- Support a phased IT load growth path from 1 MW to 4 MW without disruption to live production systems.
- Achieve an annualized PUE of ≤ 1.35 and a Water Usage Effectiveness (WUE) of ≤ 1.0 L/kWh.
- Establish a converged, spine-leaf network fabric supporting 100/400GbE core connectivity with clear growth to 800GbE.
- Implement a defense-in-depth security architecture spanning physical access control, network segmentation, and Zero Trust identity controls.
- Deliver the facility on a 18-month construction and commissioning schedule with a total capital budget envelope defined in Section 3.3.

1.2 Document Scope and Audience

This document is written primarily for data center design engineers, network architects, systems engineers, project managers, and operations staff who will design, build, and run the facility. Business case and feasibility material is included so that technical decisions can be traced back to the underlying commercial drivers, but the level of technical detail throughout the document assumes familiarity with data center engineering fundamentals.

1.3 How to Use This Document

Part I establishes why the Project exists and the constraints it must satisfy. Part II defines the physical building and its power, cooling, fire, and physical security systems. Part III defines the IT payload: compute, storage, network, and cabling, including rack-level design standards. Part IV defines how the Project will be delivered, including schedule, budget, procurement, and risk management. Part V defines how the facility will be operated and maintained once live. Part VI contains reference material including a glossary, standards catalogue, and checklists used throughout construction, commissioning, and operations.

2. Project Background and Objectives

2.1 Background

Growth in application workloads, data retention requirements, and the shift toward hybrid cloud architectures has driven demand for additional, dedicated data center capacity beyond what existing facilities or colocation arrangements can economically support. The Project is chartered to design and build a purpose-built facility that gives the organization direct control over capacity planning, security posture, and total cost of ownership, while reducing dependency on third-party colocation providers for critical workloads.

The facility will host a mix of virtualized compute, containerized platforms, database and storage tiers, and network edge services. It will be designed from the outset to support both traditional three-tier applications and cloud-native, API-driven workloads, with connectivity to public cloud providers via dedicated interconnects.

2.2 Strategic Drivers

- Consolidation of workloads currently spread across multiple legacy sites and colocation cages.
- Reduction in recurring colocation and cross-connect costs over a 10-year total cost of ownership horizon.
- Improved control over physical and network security posture for regulated workloads.
- Capacity headroom to support projected compute and storage growth of 25-30% year over year for five years.
- Improved sustainability metrics (PUE, renewable energy sourcing, water usage) versus existing facilities.

2.3 Project Objectives

The Project's objectives are grouped into four categories: availability, capacity, efficiency, and security. Availability objectives target Tier III-equivalent concurrent maintainability with no single point of failure in the critical power or cooling chain. Capacity objectives define a phased build-out path so capital is deployed in step with demand rather than as a single large up-front investment. Efficiency objectives target industry-leading PUE and WUE figures consistent with current ASHRAE and Uptime Institute guidance. Security objectives require a layered physical and logical security model meeting the requirements described in Part III, Section 17.

2.4 Success Criteria

Category	Metric	Target
Availability	Annual uptime	≥ 99.982% (Tier III)
Availability	Unplanned outage MTTR	< 4 hours
Efficiency	Design PUE (annualized)	≤ 1.35

Category	Metric	Target
Efficiency	Water Usage Effectiveness	≤ 1.0 L/kWh
Capacity	Initial critical IT load	1 MW (Phase 1)
Capacity	Build-out critical IT load	4 MW (Phase 4)
Schedule	Design-to-commissioning	18 months
Security	Physical security incidents	Zero tolerance for unauthorized access

3. Business Case and Feasibility Study

3.1 Market and Demand Analysis

Internal workload forecasting projects sustained compute and storage growth driven by three primary demand sources: core transactional application growth, expansion of data analytics and machine learning platforms, and data retention requirements introduced by regulatory and contractual obligations. Demand modeling over a 10-year horizon indicates a requirement for approximately 3,800 to 4,200 kW of critical IT load by year eight, which forms the basis for the Project's 4 MW build-out ceiling.

A build-versus-buy analysis compared a dedicated facility against continued expansion within third-party colocation. Colocation offers faster time-to-capacity but carries a materially higher 10-year total cost of ownership once recurring power, space, and cross-connect fees are modeled against a fully loaded owned-facility cost, particularly once utilization exceeds roughly 1.5 MW of sustained critical load.

3.2 Site Selection Criteria

Site selection weighed power availability and cost, fiber connectivity and carrier diversity, seismic and flood risk, proximity to a qualified labor market, and local tax and incentive structures. Sites were scored against a weighted matrix covering these factors, with power availability and network diversity weighted most heavily given their direct impact on both Tier rating achievability and long-term operating cost.

Criterion	Weight	Rationale
Utility power capacity & redundancy	25%	Two independent utility feeds required for 2N electrical topology
Fiber/carrier diversity	20%	Minimum three diverse carriers with physically separate entry points
Seismic / flood / weather risk	15%	Site must fall outside 500-year flood plain, low seismic zone
Land cost & expandability	15%	Site must support Phase 1-4 footprint plus 20% contingency
Labor market & proximity	10%	Access to qualified electrical, mechanical, and network technicians
Tax incentives / utility rate structure	10%	Favorable data center tax abatement and industrial power rates
Environmental permitting complexity	5%	Lower permitting risk shortens schedule

3.3 Financial Analysis

The financial case is built on a capital expenditure (CapEx) and operating expenditure (OpEx) model spanning design, construction, and ten years of operation. CapEx is organized by discipline: shell and civil works, electrical infrastructure, mechanical and cooling infrastructure, fire and life safety, physical security, and IT/network infrastructure. OpEx is modeled on utility power at the projected industrial rate, maintenance contracts, staffing, and consumables.

3.3.1 Capital Cost Summary (Illustrative, Phase 1)

Cost Category	% of Phase 1 CapEx
Shell, civil, and structural	22%
Electrical infrastructure (utility, UPS, generators, switchgear)	30%
Mechanical infrastructure (cooling, chillers, CRAH)	20%
Fire detection and suppression	4%
Physical security systems	3%
IT and network infrastructure (Phase 1 payload)	16%
Design, commissioning, and project management fees	5%

3.3.2 Total Cost of Ownership

A 10-year TCO model compares the owned-facility option against continued colocation expansion. The owned facility carries a higher initial capital outlay but a materially lower blended cost per kW over the 10-year horizon once utilization surpasses approximately 40% of Phase 2 capacity, driven by the elimination of colocation space and power markup, more favorable industrial power rates, and amortization of shared infrastructure across a larger footprint.

3.3.3 Return on Investment

The base case projects payback of the incremental capital investment (relative to continued colocation) within approximately 5.5 years, with sensitivity analysis showing payback ranging from 4.2 to 7.1 years depending on utility power cost trends and the pace of workload migration.

3.4 Risk Assessment (Business Case Level)

Risk	Likelihood	Impact	Mitigation
Utility power delivery delay	Medium	High	Early utility engagement, interim generator capacity
Construction cost escalation	Medium	Medium	Fixed-price contracts, contingency reserve (Section 21)
Migration schedule slippage	Medium	Medium	Phased migration plan with rollback criteria
Underutilization of built capacity	Low	Medium	Phased build-out tied to demand triggers
Regulatory/permitting delay	Low	High	Early permitting engagement, site selection weighting

4. Scope and Requirements

4.1 In-Scope

- Design, construction, and commissioning of the facility shell and all critical infrastructure (electrical, mechanical, fire, security).
- Design and deployment of Phase 1 IT and network infrastructure, including compute, storage, and network fabric.
- Data Center Infrastructure Management (DCIM) platform deployment.
- Migration planning for initial workloads (execution tracked under a separate migration workstream referencing this document).
- Operations and maintenance program design, staffing plan, and standard operating procedures.

4.2 Out-of-Scope

- Application-level architecture and software development.
- End-user computing and office IT infrastructure at the site.
- Long-term facilities lease/ownership legal structuring (covered under separate legal workstream).

4.3 Functional Requirements

Functional requirements define what the facility and its systems must do. These include supporting concurrent maintenance of any single electrical or mechanical component without service interruption, supporting rack power densities from 4 kW to 30 kW per cabinet to accommodate mixed general-purpose and high-density (GPU/AI) compute, and providing structured cabling supporting both 10/25/100GbE copper and fiber connectivity to the rack.

4.4 Non-Functional Requirements

Requirement	Target
Availability (Tier rating)	Uptime Institute Tier III (Concurrently Maintainable)
Design PUE (annualized)	≤ 1.35
Structured cabling standard	TIA-942 Rated-3
Network core bandwidth	100/400GbE spine, growth path to 800GbE
Physical security	Layered perimeter, mantrap, biometric access, 24/7 CCTV
Fire suppression	Clean-agent (e.g., FM-200/Novec 1230) with VESDA early detection
Seismic design	Per local building code plus data center equipment anchoring standards

4.5 Compliance and Standards

The facility design and operating model are aligned to a defined set of industry standards and, where applicable, regulatory frameworks. These standards are referenced throughout this document rather than restated in full; the complete list with edition references appears in Appendix B.

- Uptime Institute Tier Standard: Topology and Operational Sustainability
- TIA-942 Telecommunications Infrastructure Standard for Data Centers
- ASHRAE TC 9.9 Thermal Guidelines for Data Processing Environments
- ISO/IEC 27001 Information Security Management
- ISO 50001 Energy Management Systems
- NFPA 75 / NFPA 76 Fire Protection for Information Technology Equipment
- BICSI 002 Data Center Design and Implementation Best Practices

4A. Detailed Functional and Operational Requirements

4A.1 Facility-Level Functional Requirements

Beyond the high-level functional requirements in Section 4.3, the facility must satisfy a detailed set of operational requirements that trace directly into the design decisions made in Part II and Part III. These requirements are maintained as a formal requirements traceability matrix in the project document management system; the summary below captures the categories and representative requirements from that matrix.

ID	Requirement	Traces To
FR-001	Any single electrical component may be removed from service without IT load impact	Section 6.1
FR-002	Any single mechanical component may be removed from service without exceeding thermal envelope	Section 7.3
FR-003	Facility supports rack densities from 4 kW to 30+ kW within the same data hall	Section 6.4.1, 7.1
FR-004	Facility supports phased IT capacity growth without disruption to live racks	Section 5.3.1, 20.4
FR-005	All physical access is logged and attributable to an individual	Section 9.3
FR-006	Network fabric supports non-blocking east-west bandwidth at Phase 1 scale	Section 13.1
FR-007	Facility supports concurrent Tier III maintainability certification	Part II, Section 20.3
FR-008	All critical telemetry is retained for a minimum of 13 months for trend analysis	Section 16, 29

4A.2 Environmental and Regulatory Requirements

The facility design accounts for local building codes, environmental permitting requirements (stormwater management, generator emissions permitting, fuel storage permitting), and, where the facility will process regulated data classes, the additional physical and logical control requirements those classifications impose (for example, dedicated caged or walled enclosures for regulated workloads, enhanced audit logging, and restricted personnel access lists).

4A.3 Interoperability Requirements

Because the facility is designed to support hybrid-cloud workloads (Section 2.1), network and security architecture must interoperate cleanly with major public cloud provider interconnect services, identity

federation standards, and common infrastructure-as-code tooling used to manage both on-premises and cloud resources from a common control plane where practical.

4B. Alternatives Analysis

4B.1 Alternatives Considered

Three primary alternatives were evaluated against the do-nothing baseline of continued colocation expansion: (1) a full-build owned facility as described in this document, (2) a hybrid model combining a smaller owned facility for Tier 1 critical workloads with continued colocation for Tier 2/3 workloads, and (3) a build-to-suit lease arrangement with a third-party developer, where the facility is purpose-built to specification but owned and operated by a third party under a long-term lease.

Alternative	CapEx Profile	Control	10-Year TCO (relative)	Decision
Continued colocation expansion	Low upfront, high recurring	Low	Highest	Rejected - baseline
Full-build owned facility	High upfront	High	Lowest at scale	Selected
Hybrid owned + colocation	Medium upfront	Medium	Medium	Rejected - complexity
Build-to-suit lease	Low upfront, high recurring lease	Medium	Medium-High	Rejected - control

4B.2 Rationale for Selected Alternative

The full-build owned facility was selected because it delivers the lowest 10-year total cost of ownership once utilization crosses the threshold identified in Section 3.3.2, provides full control over security posture for regulated workloads, and avoids the recurring escalation risk inherent in both colocation and build-to-suit lease pricing models. The higher upfront capital requirement is mitigated by the phased build-out strategy (Section 1.1, Section 20.4), which aligns capital deployment with demonstrated demand.

4C. Stakeholder Analysis

Stakeholder Group	Interest	Engagement Approach
Executive sponsors	ROI, strategic alignment, risk exposure	Monthly Steering Committee reporting
IT/Network engineering	Technical feasibility, operability	Direct participation in design (Part III)
Security/compliance	Regulatory alignment, risk posture	Security design review gates
Finance	Budget adherence, TCO accuracy	Monthly cost reporting (Section 21.2)
Facilities/real estate	Site logistics, lease/ownership terms	Site selection and civil works coordination
Future operations staff	Operability, staffing, safety	Operations readiness workstream (Part V)

4D. Assumptions and Constraints

4D.1 Key Assumptions

- Utility power capacity and a second diverse feed can be secured at the selected site within the Section 20 schedule.
- Long-lead equipment (generators, chillers, switchgear, UPS) can be contracted on fixed-price, fixed-schedule terms consistent with Section 22.1.
- Regulatory and environmental permitting can be completed within the Stage 1-2 schedule window (Section 20.1) given early engagement.
- Workload growth follows the demand model in Section 3.1 within a reasonable variance band; material deviation triggers a re-baseline of the phased build-out plan.

4D.2 Constraints

- Total Phase 1 capital budget is fixed at the envelope defined during Steering Committee budget approval; scope changes beyond contingency require Steering Committee approval (Section 26.1).
- Schedule constraint: Phase 1 must reach operational readiness within 18 months of design freeze to align with the workload migration plan referenced in Section 4.2.
- Site constraint: the selected site's power and land constraints define the practical ceiling for the ultimate build-out (Section 3.1).

3A. Extended Market and Demand Analysis

3A.1 Demand by Workload Category

Workload Category	Current Share of Critical Load	Projected Share (Year 5)	Growth Driver
Core transactional applications	45%	35%	Steady growth, increasing efficiency per workload
Data analytics / data warehouse	20%	22%	Expanding reporting and BI use cases
AI/ML training and inferencing	8%	20%	New capability build-out, GPU-dense growth
Storage / backup / archive	17%	15%	Data retention growth, offset by efficiency gains
Network/edge/security appliances	10%	8%	Relatively stable, scales sub-linearly with compute

3A.2 Comparable Facility Benchmarking

Design targets in Section 4.4 were benchmarked against publicly reported figures for comparable-scale enterprise facilities, informing the ≤ 1.35 PUE and Tier III availability targets as achievable, industry-competitive figures rather than aspirational outliers, while remaining consistent with the budget envelope in Section 3.3.

3A.3 Sensitivity Analysis on Demand Forecast

Scenario	Year 5 Critical Load	Implication
Low growth (15%/yr)	~2.0 MW	Phase 3 trigger delayed, Phase 2 sufficient through year 5
Base case (25-30%/yr)	~3.2-3.5 MW	Phase 3 triggered ~year 4, consistent with base plan
High growth (40%/yr, AI-driven)	~4.5 MW+	Phase 4 triggered early; site expansion ceiling re-evaluated

3B. Extended Financial Model Detail

3B.1 Cost of Capital and Discounting

The financial model discounts future cash flows using the organization's weighted average cost of capital, consistent with standard capital project evaluation practice; the specific discount rate, along with the full cash flow model, is maintained in the finance-owned business case workbook referenced from this document rather than reproduced here in full, given it is subject to periodic treasury updates independent of the technical design.

3B.2 OpEx Model Detail

OpEx Category	Approx. % of Annual OpEx (Phase 1, steady state)
Utility power	45%
Maintenance contracts (electrical, mechanical, fire, security)	20%
Staffing (facility ops, NOC, security)	22%
Consumables and minor capital replacement	6%
Software/DCIM/monitoring licensing	4%
Insurance and other	3%

3B.3 Sensitivity of TCO to Utility Rate

Given utility power represents the largest single OpEx category, the TCO model is most sensitive to the industrial power rate assumption used in Section 3.2; a sensitivity band of $\pm 15\%$ on the assumed rate was applied in the base case ROI range reported in Section 3.3.3, and rate risk is tracked as a standing item in the business-case-level risk register (Section 3.4).

3C. Extended Compliance and Regulatory Context

3C.1 Data Classification and Handling

Workloads hosted at the facility are classified per the organization's data classification policy (public, internal, confidential, restricted), with the classification driving specific control requirements referenced throughout this document: encryption requirements (Section 17.3), physical access restrictions (Section 9D.2), log retention periods (Section 17I.2), and backup/DR treatment (Section 12.4, Section 32.2).

3C.2 Regulatory Landscape Considerations

Depending on the specific regulated data types ultimately hosted at the facility, additional sector-specific requirements (financial services, healthcare, or regional data protection regulation, as applicable) may impose further controls beyond the baseline design in this document; a dedicated compliance gap assessment against the applicable regulatory framework is conducted as part of the security design review gates referenced in Section 4C.

2A. Organizational Change Considerations

2A.1 Impact on Existing Teams

Delivery of the Project affects existing IT operations, network engineering, and facilities teams beyond the dedicated project delivery organization (Section 18.2): existing colocation-focused processes must be adapted for an owned-facility operating model, and staff currently supporting colocation vendor relationships transition toward the direct operations roles described in Part V.

2A.2 Workforce Planning

Workforce planning for the new facility (Section 21A.2, Section 27) considers both new hires and internal transfers from existing colocation-support roles, with a defined transition timeline so institutional knowledge of current production workloads is retained through the migration described in Section 4.2.

2B. Migration Strategy Overview

2B.1 Migration Approach

While detailed migration execution is tracked under a separate migration workstream (Section 4.2), the facility design in this document is informed by the intended migration approach: a wave-based migration prioritizing non-critical workloads first to validate the facility's operational readiness before Tier 1 critical workloads are migrated, with each wave gated by a defined set of operational readiness criteria consistent with Appendix K.

2B.2 Rollback Criteria

Each migration wave defines rollback criteria (e.g., sustained performance degradation, unresolved P1 incidents attributable to the new facility) under which the wave's workloads can be reverted to their prior hosting location, providing a safety net during the transition period until the new facility has an established operational track record.

3D. Insurance and Risk Transfer

The facility's insurance program (property, business interruption, cyber liability) is structured in coordination with the risk assessment in Section 3.4 and Section 9D.1, with coverage levels and deductibles reviewed against the financial model in Section 3.3 so that residual risk not otherwise mitigated or contractually transferred (Section 23B.1) is appropriately covered.

3E. Post-Implementation Review Plan

A formal post-implementation review is scheduled for 12 months after go-live, comparing actual availability, PUE/WUE, cost, and utilization against the business case targets in Section 1.2 and Section 3, with findings feeding both the lessons-learned archive (Section 26A) and the ongoing phase-trigger and capacity planning process (Section 20.4, Section 33).

3F. Extended Site Comparison Scorecard (Illustrative)

Site Candidate	Power Score	Fiber Score	Risk Score	Cost Score	Weighted Total
Site A (selected)	9/10	9/10	8/10	7/10	8.4/10
Site B	7/10	8/10	7/10	8/10	7.4/10
Site C	8/10	6/10	6/10	9/10	7.1/10
Site D	6/10	7/10	9/10	6/10	6.9/10

3G. Extended Financial Sensitivity Table

Variable	Base Case	Downside Case	Upside Case
Utility power rate escalation (annual)	3%	5%	1.5%
Workload growth rate (annual)	27%	15%	40%
Construction cost escalation	Fixed-price (Section 22.1)	Fixed-price (Section 22.1)	Fixed-price (Section 22.1)
Payback period (years)	5.5	7.1	4.2
10-year TCO advantage vs. colocation	Base case per Section 3.3.2	Reduced, still favorable above utilization threshold	Materially improved

4E. Extended Requirements Traceability - Business and Operational

ID	Requirement	Traces To
BR-001	10-year TCO must be favorable vs. continued colocation above the utilization threshold in Section 3.3.2	Section 3.3.2, 3B
BR-002	Payback period target of 4-7 years under base and sensitivity cases	Section 3.3.3, 3G
OR-001	Operations staffing model must support 24x7x365 coverage from go-live	Section 27
OR-002	Migration plan must support wave-based, reversible migration	Section 2B
OR-003	Post-implementation review conducted at 12 months	Section 3E
SR-001	Facility must support future regulatory compliance frameworks without structural redesign	Section 3C.2, 4A.2

5. Site and Building Design

5.1 Site Layout and Civil Works

The site is organized into three functional zones: the technical building housing white space and gray space, the yard housing generators, fuel storage, and outdoor mechanical plant, and the perimeter zone housing vehicle screening, security fencing, and setback landscaping. Civil works include site grading to direct stormwater away from the building envelope, a reinforced concrete slab rated for raised-floor and containment loading, and dedicated utility entrance vaults for the two independent utility feeds.

Vehicle access is separated into a screened delivery/loading area for generator fuel and equipment deliveries, and a separate staff and visitor entrance routed through the security vestibule described in Section 9. Setback distances from the perimeter fence to the building follow CPTED (Crime Prevention Through Environmental Design) principles to eliminate concealment points and provide standoff distance against vehicle-borne threats.

5.2 Facility Layout: White Space and Gray Space

White space refers to the raised-floor or slab data hall area housing IT racks, in-row cooling, and PDUs. Gray space refers to the supporting mechanical, electrical, and fire suppression plant rooms, including the UPS room, battery room, switchgear room, and mechanical plant room. The Phase 1 layout allocates approximately 1,100 square meters of white space and 900 square meters of gray space, with the white space subdivided into four data halls of roughly 275 square meters each to support fire compartmentalization and phased fit-out.

5.2.1 Data Hall Layout Principles

- Hot aisle / cold aisle orientation with rows aligned to the air handling units for shortest supply air path.
- Containment (cold-aisle or hot-aisle, per Section 7.2) on all production rows from Phase 1 onward.
- Minimum 1.2 m clearance in cold aisles and 1.0 m in hot aisles for maintenance access and code compliance.
- Structural floor loading rated for a minimum of 1,200 kg/m² to support high-density rack rows.
- Dedicated network meet-me room adjacent to the carrier entrance facility, isolated from general white space.

5.3 Structural Design

The building structure is a tilt-up or precast concrete shell (final selection per site-specific engineering) with a raised access floor system in the white space rated for the point loads associated with fully populated high-density cabinets. Roof structure is designed to support rooftop mechanical plant (condensers, dry coolers, or cooling towers per the mechanical design selected in Section 7) with appropriate vibration isolation and acoustic screening.

5.3.1 Expansion and Phasing Strategy

The building shell for the full 4 MW build-out is constructed in Phase 1 to avoid disruptive future construction adjacent to a live data hall. Interior fit-out of data halls 2 through 4, along with the corresponding electrical and mechanical plant, is deployed incrementally as Phases 2 through 4 are triggered by utilization thresholds defined in Section 20.

6. Electrical Infrastructure

6.1 Utility Power and Redundancy Architecture

The facility is served by two physically diverse utility feeds entering the site at separate points, each sized to independently carry the full critical load at ultimate build-out. This dual-feed arrangement is a prerequisite for the 2N electrical topology on the critical UPS and distribution path and underpins the Tier III concurrent maintainability objective.

Electrical distribution follows an N+1 topology at the mechanical support level and a 2N topology on the critical UPS, battery, and PDU path serving IT racks. Each rack receives dual, independently sourced power feeds (A and B) from separate UPS systems, separate PDUs, and separate upstream switchgear lineups, allowing any single component in either path to be taken out of service for maintenance without impacting IT load.

6.1.1 Redundancy Levels by System

System	Redundancy Level	Notes
Utility feed	2 independent feeds	Physically diverse entry points
Standby generators	N+1	Sized for full critical + mechanical load
UPS systems	2N	Dual independent A/B UPS paths
Battery / energy storage	2N	10-15 minute ride-through at full load
Switchgear (medium voltage)	N+1	Main-tie-main configuration
PDUs / RPPs	2N	Dual-corded IT load, A/B feeds
Static transfer switches	N+1	Where single-corded loads require it

6.2 UPS Systems

Uninterruptible power supply capacity is provided by modular, double-conversion UPS systems configured in a 2N architecture, with each of the A and B paths independently capable of carrying full critical load. UPS sizing includes headroom for the ultimate 4 MW build-out, with modules added incrementally in step with IT load growth so that capital is not stranded ahead of demand. Battery strings (or equivalent flywheel/lithium-ion energy storage, per final vendor selection) are sized to provide a minimum 10-15 minute ride-through at full rated load, sufficient to bridge to generator start and stable output.

6.3 Standby Generators and Fuel Systems

Standby power is provided by diesel generators in an N+1 configuration, each sized to carry the full facility critical and mechanical load independently. On-site fuel storage is sized for a minimum of 72 hours of full-load runtime, with a service contract in place for guaranteed fuel resupply within 24 hours during extended outages. Generators are equipped with automatic transfer switches, block heaters for cold-start reliability, and are load-bank tested on a defined maintenance cadence described in Part V, Section 30.

6.4 Switchgear, PDUs, and Power Distribution

Medium-voltage switchgear steps utility power down for on-site distribution, with a main-tie-main configuration allowing either utility feed to serve the full load if the other is unavailable. Power then passes through UPS systems to Power Distribution Units (PDUs) and Remote Power Panels (RPPs) at the row or rack level. Rack Power Distribution Units (rPDUs) are metered and, where required by rack density, are intelligent (network-monitored) units supporting per-outlet power measurement, feeding into the DCIM platform described in Section 16.

6.4.1 Rack-Level Power Design

Rack Class	Typical Density	Power Feed
Standard compute/network	4-8 kW	Dual 1-phase or 3-phase, A/B feed
High-density compute	10-20 kW	Dual 3-phase, A/B feed
GPU / AI training racks	20-30+ kW	Dual 3-phase high-amperage, liquid-cooling ready

6.5 Grounding, Bonding, and Power Quality

The facility implements a common bonding network per TIA-942 and IEEE recommendations, tying structural steel, rack frames, cable trays, and electrical equipment to a single reference ground grid to minimize ground potential differences and support surge protection. Power quality is monitored continuously through the DCIM platform, with harmonic distortion, voltage sag/swell, and transient events logged and alarmed.

7. Mechanical and Cooling Infrastructure

7.1 Cooling Architecture

The Phase 1 cooling design uses a chilled-water plant feeding Computer Room Air Handler (CRAH) units at the perimeter of each data hall, supplemented by in-row cooling units in high-density zones. This hybrid approach allows standard-density rows (up to roughly 8 kW/rack) to be served efficiently by perimeter CRAH units, while high-density and GPU rows are served by in-row or rear-door heat exchanger units capable of higher sensible heat removal per rack, with a defined migration path to direct liquid cooling (cold-plate) for future GPU generations exceeding air-cooling limits.

7.1.1 Chilled Water Plant

The central plant consists of N+1 chillers, redundant primary and secondary chilled-water pumps, and either cooling towers or air-cooled condensers depending on final site water availability analysis. The plant is designed to take advantage of free/economizer cooling during favorable ambient conditions, reducing chiller run-hours and directly improving annualized PUE.

7.2 Airflow Management and Containment

All production data halls implement cold-aisle containment from Phase 1, using containment doors and roof panels to physically separate supply (cold) and return (hot) air paths. This prevents bypass and recirculation air, allows CRAH/in-row set points to be raised safely (per ASHRAE TC 9.9 recommended and allowable envelopes), and materially improves cooling unit efficiency. Blanking panels are mandatory in all unused rack U-space, and brush grommets are required at all cable pass-throughs to prevent air leakage.

7.2.1 ASHRAE Thermal Envelope Targets

Class	Recommended Range	Allowable Range
A1/A2 (typical enterprise IT)	18-27°C	15-32°C
Humidity	5.5°C DP to 60% RH / 15°C DP	20-80% RH non-condensing

7.3 Redundancy and Maintainability

Cooling infrastructure follows the same concurrent-maintainability principle as electrical infrastructure: N+1 chillers and pumps, N+1 CRAH units per data hall (sized so that any single unit can be taken offline without exceeding the design thermal envelope), and dual chilled-water supply/return loops feeding each data hall where the Tier objective requires it.

7.4 PUE, WUE, and Efficiency Monitoring

Power Usage Effectiveness and Water Usage Effectiveness are calculated continuously from metered inputs (utility power at the meter, IT load at the PDU, water consumption at the plant) and reported through the DCIM platform. Efficiency targets are reviewed quarterly against the ≤ 1.35 PUE and ≤ 1.0 L/kWh WUE objectives set in Section 4.4, with corrective actions tracked when a data hall trends outside target for two consecutive quarters.

8. Fire Detection and Suppression

8.1 Detection Systems

Very Early Smoke Detection Apparatus (VESDA) aspirating smoke detection is deployed above and below the raised floor (where applicable) and within air handling return paths in every data hall, providing detection well ahead of the threshold required to trigger conventional point smoke detectors. Detection zones are cross-referenced with the fire suppression release logic to prevent false-positive discharge from a single-zone alarm.

8.2 Suppression Systems

Data halls are protected by clean-agent fire suppression (e.g., FM-200, Novec 1230, or equivalent per final environmental and code review) rather than water-based sprinklers, to eliminate the risk of water damage to IT equipment during a suppression event. Suppression is designed as a cross-zoned system requiring detection from two independent detection circuits before automatic discharge, with manual abort and manual release stations at all data hall entrances per NFPA 2001.

Gray-space areas such as loading docks and office space use conventional pre-action sprinkler systems per NFPA 13 and local code, isolated from the clean-agent zones.

8.3 Compliance and Testing

Fire detection and suppression systems are commissioned per NFPA 75/76 and are subject to the inspection, testing, and maintenance cadence defined in Part V, Section 30, including annual full-discharge testing on a rotating schedule and monthly detector sensitivity verification.

9. Physical Security

9.1 Layered Security Model

Physical security follows a layered (defense-in-depth) model with four concentric layers: perimeter (fencing, vehicle barriers, lighting), building envelope (controlled entrances, intrusion detection), interior circulation (badge-controlled corridors), and the data hall itself (mantrap with biometric verification). Each layer requires successful authentication at the prior layer, and no single credential type is sufficient to reach the data hall unescorted.

9.2 Perimeter Security

- Anti-ram perimeter fencing and vehicle barriers rated to the threat level identified in the site risk assessment.
- Controlled vehicle sally port for deliveries, with under-vehicle inspection where risk assessment warrants it.
- CPTED-aligned lighting and landscaping eliminating concealment points along the fence line.

9.3 Access Control and Credentialing

Access control uses multi-factor credentials (badge plus PIN or biometric) at all controlled doors, with role-based access provisioning tied to the identity and access management system described in Section 17.4. All access events are logged, retained per the retention schedule in Appendix B, and reviewed as part of the quarterly access recertification process described in Part V.

9.4 Data Hall Entry: Mantrap and Biometrics

Entry to each data hall is controlled through a mantrap (interlocked double-door vestibule) requiring biometric verification (e.g., fingerprint or iris, per final vendor selection) in addition to badge credentials. The mantrap enforces single-person passage and logs both successful and failed entry attempts to the security information and event management (SIEM) platform referenced in Section 17.

9.5 Surveillance

CCTV coverage provides overlapping views of all perimeter approaches, building entrances, interior corridors, and data hall aisles, recorded continuously with a minimum 90-day retention period (or longer where regulatory requirements apply). Camera feeds are monitored 24/7 by the on-site security operations function described in Part V, Section 27, with automated analytics (e.g., loitering, tailgating detection) feeding alerts into the security monitoring console.

9A. Detailed Electrical Design Specifications

9A.1 Medium Voltage Distribution

Utility power is received at medium voltage (typical range 13.8kV-34.5kV depending on local utility standards) at a dedicated utility substation on-site, stepped down through unit substations feeding the low-voltage switchgear that serves UPS input and mechanical plant. Medium-voltage switchgear is configured in a main-tie-main arrangement so either utility feed can serve the full connected load if the other feed or its associated switchgear requires maintenance or experiences a fault.

9A.2 UPS Topology Detail

Each data hall is served by a minimum of two independent UPS systems (the A and B paths), each sized with N+1 module redundancy internally in addition to the 2N path-level redundancy. This double layer of redundancy (N+1 within each path, 2N across paths) ensures that a single UPS module failure on the A path does not force a full transfer of A-path load onto the B path, preserving true concurrent maintainability even during a component fault.

9A.2.1 UPS Sizing Reference Table

Phase	Critical IT Load	UPS Capacity per Path (N+1 modules)	Runtime at Full Load
Phase 1	1.0 MW	1.25 MW (5x 300kW modules, N+1)	12 minutes
Phase 2	2.0 MW	2.5 MW (10x 300kW modules, N+1)	12 minutes
Phase 3	3.0 MW	3.75 MW (15x 300kW modules, N+1)	12 minutes
Phase 4	4.0 MW	5.0 MW (20x 300kW modules, N+1)	12 minutes

9A.3 Generator Plant Detail

Standby generation is provided by a plant of diesel generators sized so that N units carry full critical plus mechanical load, with one additional unit (N+1) providing redundancy for maintenance or unit failure. Generators start automatically on loss of utility power, reaching stable rated output within the design threshold (typically 10-15 seconds) so that UPS battery/energy-storage ride-through bridges the gap without any interruption to IT load.

9A.3.1 Fuel System Design

Bulk fuel storage is sized for 72 hours of full-load runtime across the generator plant, held in double-walled, leak-monitored tanks meeting local environmental code. A day tank at each generator provides local fuel buffering, automatically replenished from the bulk storage via redundant fuel transfer pumps. Fuel quality is maintained through a scheduled polishing/filtration program (Section 30) to prevent

microbial growth and particulate contamination that could compromise generator reliability during an extended outage.

9A.4 Static Transfer Switches and Single-Corded Loads

While the design standard requires dual-corded IT equipment, certain legacy or specialized single-corded devices may need to be accommodated. Where this is unavoidable, Static Transfer Switches (STS) are used to automatically switch a single-corded load between the A and B feeds within a sub-cycle transfer time, preserving effective redundancy without requiring the device itself to support dual power inputs.

9A.5 Emergency Power Off (EPO) and Life Safety Interlocks

EPO controls are provided per local code at data hall entrances, wired to shut down power to IT equipment (but not life-safety systems such as emergency lighting) in the event of an immediate life-safety emergency. EPO circuits are physically guarded to prevent accidental activation and are tested as part of the commissioning program (Section 20.3) and periodic life-safety testing (Section 30).

9A.6 Lightning and Surge Protection

The facility implements a lightning protection system per applicable code (e.g., NFPA 780 or local equivalent), with surge protective devices (SPDs) installed at the service entrance, at UPS input/output, and at critical distribution panels to protect sensitive electronic equipment from transient overvoltage events.

9B. Detailed Mechanical Design Specifications

9B.1 Chiller Plant Detail

Phase	Critical IT Load	Cooling Capacity Required (approx.)	Chiller Configuration
Phase 1	1.0 MW	~350 tons	3x chillers (N+1, 2 duty + 1 standby)
Phase 2	2.0 MW	~700 tons	4x chillers (N+1)
Phase 3	3.0 MW	~1,050 tons	5x chillers (N+1)
Phase 4	4.0 MW	~1,400 tons	6x chillers (N+1)

9B.2 Economizer and Free Cooling Strategy

The mechanical design incorporates air-side or water-side economization (per final site climate analysis) to reduce mechanical cooling run-hours whenever ambient conditions allow the chilled-water loop, or in some configurations outside air directly, to meet the data hall's cooling load without full chiller operation. Climate analysis for the selected site is used to project annual free-cooling hours, which feeds directly into the annualized PUE model in Section 7.4.

9B.3 Humidity Control

Humidification and dehumidification are provided at the air handling unit level, controlled to maintain the ASHRAE allowable envelope defined in Section 7.2.1 across all data halls. Humidity control is coordinated facility-wide through the BMS to avoid the common failure mode where adjacent units simultaneously humidify and dehumidify against each other, wasting energy without net environmental benefit.

9B.4 Leak Detection

Water leak detection cabling is installed beneath raised floor areas and along chilled-water piping runs throughout gray-space mechanical rooms, with alarms integrated into the DCIM/BMS platform (Section 16) to enable rapid isolation of any water intrusion before it can reach IT equipment.

9C. Detailed Fire and Life Safety Design

9C.1 Suppression Agent Selection

Clean-agent selection (e.g., FM-200, Novec 1230, or an inert gas agent such as Inergen) is finalized based on room volume, agent concentration requirements, occupant safety margins, and environmental/regulatory considerations for the selected jurisdiction. Agent quantity is calculated per NFPA 2001 design concentration requirements for each protected volume, with reserve agent cylinders held on-site for rapid system recharge after a discharge event.

9C.2 Egress and Life Safety

Data hall layouts maintain code-compliant egress path widths and travel distances, with emergency lighting and illuminated exit signage on a dedicated life-safety circuit independent of the general UPS-backed IT power path, ensuring egress lighting remains available even during a full IT power event.

9C.3 Smoke Control and Compartmentalization

Data halls are fire-compartmentalized from each other and from gray-space mechanical/electrical rooms using rated fire/smoke barriers, limiting the potential spread of a fire or smoke event to a single data hall and supporting the phased fit-out strategy described in Section 5.3.1.

9D. Detailed Physical Security Design

9D.1 Security Risk Assessment Methodology

The physical security design is derived from a formal threat and vulnerability risk assessment considering the facility's asset criticality, the local threat environment, and applicable regulatory requirements for any regulated workloads hosted at the site. The assessment output directly sets the perimeter hardening level (Section 9.2), the CCTV coverage and retention requirements (Section 9.5), and staffing levels for the on-site security function (Part V, Section 27).

9D.2 Visitor and Vendor Management

All visitors and vendors are pre-registered, verified against government-issued identification at the security vestibule, and escorted by authorized staff for the duration of their visit unless granted unescorted access through the formal background-check and access-provisioning process described in Section 17.4. Vendor access to specific racks or rows is restricted through the access control system to only the areas relevant to their service ticket.

9D.3 Security Incident Response

Physical security incidents (unauthorized access attempts, tailgating alerts, forced-door alarms) are handled through the same incident management process described in Part V, Section 31, with immediate on-site security officer response and, for confirmed breaches, law-enforcement notification per the facility's security incident response plan.

9D.4 Integration with IT Security

Physical access events are correlated with IT/network access events in the SIEM platform (Section 17.4.1), enabling detection of anomalies such as a badge-in event at the data hall with no corresponding expected maintenance activity, or an administrative login occurring without a corresponding physical presence for procedures that require on-site execution.

9E. Structural and Seismic Design Detail

9E.1 Floor Loading Design Criteria

Area	Design Live Load	Notes
White space (raised floor)	1,200 kg/m ² minimum	Supports fully populated high-density rows
Battery/UPS room	1,500 kg/m ² minimum	Accounts for battery string weight
Generator yard pad	Per vendor footprint + seismic factor	Independent foundation from main structure
Mechanical plant room (chillers)	1,000 kg/m ² minimum	Includes vibration isolation mounts

9E.2 Seismic Anchoring

In seismic design categories requiring it, all IT racks, UPS cabinets, battery racks, and major mechanical equipment are anchored per the site-specific seismic engineering analysis, using seismically rated rack frames and bracing kits for IT cabinets, consistent with the equipment anchoring guidance referenced in Section 4.4 (non-functional requirements).

9E.3 Wind and Weather Design

Roof-mounted mechanical equipment (condensers, dry coolers) and the building envelope are designed to the local wind load code, with equipment screening providing both acoustic and weather protection consistent with the roof structural allowance described in Section 5.3.

9F. Representative Generator Specification

Parameter	Representative Value (Phase 1 unit)
Prime mover	Diesel, Tier 4 Final (or local equivalent) emissions compliant
Standby rating	Sized to Phase 1 critical + mechanical load per unit, N+1 units installed
Start time to rated output	≤ 15 seconds
Fuel tank (day tank)	Sized for minimum 8 hours at full load
Sound attenuation enclosure	Sized to meet local noise ordinance at property line
Block heater	Included, thermostatically controlled for cold-start reliability
Paralleling switchgear	Digital paralleling controls supporting N+1 load-sharing

9G. Water Treatment and Cooling Tower Management

9G.1 Water Treatment Program

Where cooling towers are part of the final mechanical design (Section 9B.1), a water treatment program controls scale, corrosion, and biological growth (including Legionella risk management) through chemical treatment dosing, cycles-of-concentration control, and periodic bacteriological testing, documented as part of the preventive maintenance program in Section 30.

9G.2 Blowdown and Makeup Water Management

Cooling tower blowdown and makeup water rates are monitored continuously and feed the WUE calculation described in Section 7.4; the water stewardship measures described in Section 36A.2 are evaluated during detailed mechanical design to reduce net water consumption.

9H. Detailed Physical Security Hardware Schedule

System	Representative Coverage (Phase 1)
Perimeter fencing/vehicle barriers	Full site perimeter per risk assessment (Section 9D.1)
Badge readers (multi-factor)	All controlled doors: perimeter, building entrances, corridors, data hall mantraps
Biometric readers	Data hall mantraps, UPS/battery room, network meet-me room
CCTV cameras	Perimeter approaches, all entrances, all interior corridors, every data hall aisle end
Door contacts / intrusion sensors	All exterior doors and all gray-space equipment room doors
Security operations console	1 primary console in on-site SOC/NOC, redundant remote monitoring capability

9H.1 CCTV Coverage and Retention Design

Camera placement is designed for overlapping coverage with no unmonitored approach to a controlled door, at a resolution sufficient for facial identification at entrances and for general situational awareness in aisles and corridors, retained for the minimum period defined in Section 9.5 and longer where a specific data classification requires extended retention.

9I. Environmental Permitting and Regulatory Approval Process

9I.1 Permitting Workstream

Environmental and building permitting runs as a parallel workstream to design (Section 20.1), beginning with early regulatory engagement during site selection (Section 3.2) to identify permitting risk ahead of site commitment. Key permits typically include building/structural permits, generator air emissions permits, fuel storage permits, stormwater discharge permits, and, where applicable, water use permits for cooling tower makeup.

9I.2 Permitting Risk Management

Permitting risk is tracked in the project risk register (Section 23.2) with the Facility Design Lead as the accountable owner; any permitting delay with potential to impact the Section 20.2 milestone schedule is escalated to the Steering Committee per the standard escalation path in Section 18.3.

9J. Facility Signage and Wayfinding

Interior signage identifies fire compartments, EPO locations, electrical panel identifiers, and emergency egress routes consistent with the labeling conventions used in the DCIM platform (Section 14.4), so that both routine staff and emergency responders unfamiliar with the specific facility layout can navigate safely and correctly identify critical equipment during an incident.

9K. Building Management System (BMS) Detail

9K.1 BMS Architecture

The BMS provides centralized monitoring and control of mechanical plant (chillers, pumps, CRAH/in-row units, economizer dampers), electrical monitoring points (breaker status, power meters), and life-safety system status feeds, presenting a unified operator interface in the NOC alongside the DCIM and network monitoring dashboards described in Section 29. The BMS is deployed on a dedicated control-system network segment, isolated from general IT VLANs per the segmentation model in Section 13.2, to reduce the attack surface of operational technology (OT) systems.

9K.2 BMS Redundancy

BMS controllers are deployed in a redundant, hot-standby configuration so that a single controller failure does not result in loss of monitoring or automated control of mechanical plant; where a controller does fail, mechanical equipment is designed to fail to a safe default operating state (e.g., last-known-good setpoint) rather than an uncontrolled state.

9K.3 OT Cybersecurity

Building management and mechanical/electrical control systems are treated as operational technology (OT) requiring dedicated security controls beyond standard IT practice: network isolation from the general IT network, restricted vendor remote-access paths (brokered through the PAM solution referenced in Section 17.2), and change control for BMS logic changes following the same process used for network configuration (Section 17B.2.1).

9L. Elevator, Loading Dock, and Material Handling

Where the facility layout requires vertical or extended horizontal material movement (e.g., a multi-story design or extended loading dock-to-white-space distance), material handling equipment (elevators rated for IT equipment loads, pallet jacks, lift-assist devices for heavy chassis) is provisioned and included in the operations SOP library (Section 28.1) covering safe equipment delivery and removal procedures.

9M. Acoustic Design

Generator enclosures, chiller plant, and CRAH/in-row units are selected and, where necessary, supplemented with acoustic attenuation to meet both the local noise ordinance at the property line (Section 9E.3) and occupational noise exposure limits for staff working in gray-space plant rooms, with hearing protection PPE requirements defined in the EHS program (Section 27C.1) for any area exceeding the applicable exposure threshold.

9N. Landscaping and Stormwater Management

Site landscaping is designed jointly for CPTED perimeter security objectives (Section 9D.1) and stormwater management, directing site runoff to detention/retention facilities sized per the environmental permitting requirements identified in Section 9I.1, with vegetation selection avoiding species that could obscure CCTV sightlines or perimeter lighting coverage.

90. Facility Commissioning Documentation Package

90.1 As-Built Documentation Requirements

Upon substantial completion of each system, the installing contractor delivers a complete as-built documentation package (drawings, equipment schedules, control sequences, warranty documentation) reconciled against the design baseline (Section 26.2) and loaded into the DCIM/document management system before that system is accepted into the commissioning program described in Section 20.3.

90.2 Operations and Maintenance Manuals

Vendor-supplied operations and maintenance manuals for all major equipment are compiled into a facility-specific O&M manual set, cross-referenced with the internal SOP/MOP library (Section 28) so operations staff have a single, consistent reference set rather than needing to reconcile vendor documentation against internal procedures during an incident.

9P. Redundant Utility and Backup Interconnection Studies

9P.1 Utility Coordination Study

A protective device coordination study, performed jointly with the serving utility during detailed electrical design, confirms that protective relay settings across the utility interconnect, on-site switchgear, and downstream distribution correctly isolate faults at the lowest practical level, preventing a fault in one data hall's distribution from unnecessarily tripping upstream protection serving other data halls.

9P.2 Arc Flash Study

An arc flash hazard analysis is performed for all electrical distribution equipment, establishing incident energy levels and the corresponding PPE category required for work at each panel, posted at the equipment per NFPA 70E and referenced in the relevant electrical MOPs (Section 28.3, Section 27C.1).

9Q. Facility Expansion and Future-Proofing Considerations

Beyond the Phase 1-4 build-out described in Section 5.3.1, the site master plan reserves land and utility capacity headroom for a potential future expansion beyond the current 4 MW ceiling, subject to a separate future business case, so that a successful facility is not artificially constrained by decisions made at initial design time. Utility interconnect agreements are structured, where the local utility permits, to include an option for additional future capacity at a pre-negotiated cost basis.

9R. Electrical Equipment Reference Schedule (Phase 1)

Equipment	Quantity	Rating (representative)	Redundancy
Utility service transformer	2	Sized to full Phase 1-4 ultimate load	2 independent feeds
Medium-voltage switchgear lineup	2	Main-tie-main	N+1 effective
Standby generator	4 (Phase 1: 3 duty + 1 standby)	~1.25 MW each	N+1
UPS module	10 (5 per path x 2 paths)	300 kW each	2N with N+1 per path
Battery/energy storage string	10 (paired with UPS modules)	12-minute runtime at full module load	2N
Rack PDU (rPDU)	2 per rack	Per rack density class (Section 6.4.1)	A/B dual feed

9S. Mechanical Equipment Reference Schedule (Phase 1)

Equipment	Quantity	Rating (representative)	Redundancy
Chiller	3	~120 tons each	N+1 (2 duty + 1 standby)
Primary chilled water pump	3	Sized to chiller flow	N+1
Secondary chilled water pump	3	Sized to data hall demand	N+1
CRAH unit	8 per data hall	~30 tons each	N+1 per data hall
In-row cooling unit (high-density zones)	As required by rack density plan	Per vendor spec	N+1 per zone
Cooling tower / dry cooler	3	Sized to chiller heat rejection	N+1

9T. Fire and Security Equipment Reference Schedule (Phase 1)

Equipment	Quantity (Phase 1)	Notes
VESDA detection unit	1 per data hall + gray-space rooms	Cross-zoned with suppression release
Clean-agent suppression cylinder bank	1 per protected zone + reserve	Sized per NFPA 2001 concentration calc
Badge/biometric reader	Per Appendix H schedule (Section 9H)	Multi-factor at all controlled points
CCTV camera	Per coverage design (Section 9H.1)	90-day minimum retention
Mantrap vestibule	1 per data hall entrance	Fail-secure interlock

9U. Site Utility Interconnection Detail

9U.1 Utility Metering and Billing Points

Each of the two utility feeds terminates at an independent utility revenue meter, allowing the facility's OpEx model (Section 3B.2) to reconcile actual utility billing against the DCIM-calculated PUE (Section 7.4) and to detect any unexpected imbalance in loading between the A and B utility paths that could indicate a developing fault condition.

9U.2 Utility Outage History and Reliability Basis

Site selection (Section 3.2) included a review of the candidate utility's historical outage frequency and duration (SAIDI/SAIFI figures where available) for the specific feeder(s) serving the site, informing both the generator runtime sizing (Section 6.3) and the overall confidence in the availability target (Section 4.4).

9V. Detailed Containment and Airflow Verification

9V.1 Computational Fluid Dynamics (CFD) Modeling

Data hall airflow design is validated through CFD modeling during detailed mechanical design, confirming that the planned CRAH/in-row unit placement and containment configuration deliver adequate supply air to every rack position, including the high-density zones described in Section 6.4.1, before construction of containment and cooling infrastructure begins.

9V.2 Post-Installation Airflow Verification

Following installation, airflow is verified in the field at each rack position as part of Level 2/3 mechanical commissioning (Appendix I.2), confirming the CFD model's predictions against as-built conditions and identifying any localized hot spots requiring blanking panel, containment seal, or CRAH tuning corrections before the data hall is accepted for IT load.

9W. Facility Address, Wayfinding, and Emergency Responder Information

A pre-incident plan package is provided to local emergency responders (fire department, in particular) covering building layout, fire compartmentalization (Section 9C.3), clean-agent suppression zones and associated occupant safety considerations, electrical hazard locations, and designated fire department connection and command post locations, updated whenever a material change is made to the fire/life safety systems.

10. Data Center Tiering and Topology Standards

All IT and network design decisions in this Part are made against the Uptime Institute Tier III (Concurrently Maintainable) target established in Part I, and against TIA-942 Rated-3 structured cabling requirements. Concurrent maintainability at the IT layer means every active network device, storage controller, and server power path must support planned maintenance on any single component without service interruption, mirroring the electrical and mechanical redundancy defined in Part II.

10.1 Design Principles Carried Through Part III

- No single point of failure in the compute, storage, or network path serving a production workload.
- Every production server is dual-corded to independent A/B power feeds and dual-homed to independent network fabrics.
- Growth is modular: compute, storage, and network capacity are added in defined units (pods) rather than as monolithic upgrades.
- All infrastructure is instrumented and reports into the DCIM and monitoring stack described in Sections 16 and 29.

11. Server Infrastructure

11.1 Compute Architecture

Compute is delivered through a mix of 1U/2U rack-mount servers for general-purpose and storage-heavy workloads, and high-density blade or sled chassis for horizontally scaled virtualization and container hosts. GPU-dense nodes for AI/ML training workloads are deployed in dedicated high-density racks with the enhanced power and cooling design described in Sections 6.4.1 and 7.1.

11.1.1 Server Classes

Class	Form Factor	Typical Use
General purpose compute	1U/2U rack, dual-socket	Virtualization hosts, application tier
Density-optimized compute	Blade/sled chassis	Horizontally scaled container/VM hosts
Storage-dense	2U/4U, high drive-bay count	Software-defined storage nodes
GPU/AI training	4U-8U, multi-GPU	ML training, inferencing at scale
Edge/network appliance	1U	Load balancers, firewalls, WAN edge

11.2 Server Specification and Sizing Standards

Baseline server specifications are standardized into a small number of reference SKUs to simplify spares holding, firmware management, and capacity planning. Reference specifications are reviewed annually against current-generation CPU, memory, and NVMe storage options, with the standard refreshed on a 4-5 year hardware lifecycle aligned to vendor support windows.

Reference SKU	CPU	Memory	Local Storage	Network
Compute-Standard	Dual-socket, 32-64 cores/socket	512 GB - 1 TB	2x NVMe boot, RAID1	Dual 25GbE
Compute-Dense	Dual-socket, 48-64 cores/socket	1 TB	2x NVMe boot	Dual 25/100GbE
Storage-Node	Single/dual-socket	256-512 GB	12-24 bay NVMe/SAS	Dual 25/100GbE
GPU-Training	Dual-socket host + 4-8 GPU	1-2 TB	NVMe local scratch	Dual 100/400GbE (RDMA)

11.3 Virtualization and Hypervisor Architecture

The primary compute layer runs a clustered hypervisor platform (design assumes a mainstream enterprise hypervisor; final vendor selection is tracked as a procurement decision under Section 22) with clusters sized to N+1 host redundancy, allowing any single host to enter maintenance mode with automated workload evacuation. Clusters are organized by workload tier (production, non-production, DMZ) to maintain the network segmentation model described in Section 13.2.

11.3.1 Cluster Sizing Guidance

- Maximum cluster size bounded by blast-radius considerations and management-plane scaling limits, not purely by hardware capacity.
- N+1 host headroom maintained at all times to allow rolling maintenance without capacity impact.
- Resource pools and reservations enforced for latency-sensitive workloads to prevent noisy-neighbor contention.

11.4 Containerization Platform

A Kubernetes-based container platform runs alongside the virtualization layer, deployed on dedicated worker node pools sized independently from the VM estate. Control-plane components are deployed with multi-master high availability, spread across failure domains (racks/power feeds) so that loss of a single rack does not take down cluster control. Persistent storage for stateful workloads is provided through the software-defined storage layer described in Section 12, exposed via a standard container storage interface (CSI) driver.

11.4.1 Node Pool Segmentation

Node Pool	Purpose	Isolation
System/control-plane	Cluster control plane, core add-ons	Dedicated nodes, no workload scheduling
General workload	Stateless application workloads	Shared, resource-quota bound namespaces
Data/stateful	Databases, message queues	Dedicated nodes, local NVMe, anti-affinity rules
GPU workload	ML inferencing/training pods	GPU-taint isolated nodes

12. Storage Infrastructure

12.1 Storage Architecture Overview

Storage is delivered through a tiered architecture combining a high-performance all-NVMe tier for latency-sensitive databases, a general-purpose hybrid tier for standard block and file workloads, and a high-capacity object storage tier for backups, archives, and unstructured data. Storage is presented via a mix of block (Fibre Channel and iSCSI/NVMe-oF), file (NFS/SMB), and object (S3-compatible) protocols, selected per workload requirement.

12.2 SAN and NAS Architecture

The block storage layer uses a dual-fabric Storage Area Network (SAN) design: two independent Fibre Channel (or NVMe-oF, per final protocol decision) fabrics, each connected to independent SAN switches, so that a single fabric failure or planned maintenance event does not impact host connectivity. Hosts requiring block storage are dual-homed to both fabrics via multipath I/O. File storage (NAS) is delivered through clustered file storage controllers presenting NFS/SMB shares with active-active controller failover.

12.2.1 Storage Tiers and Media

Tier	Media	Typical Workload	Target Latency
Tier 0 - Performance	All-NVMe	OLTP databases, latency-critical apps	< 1 ms
Tier 1 - General Purpose	Hybrid NVMe/SSD cache + HDD capacity	General VM/app storage	1-5 ms
Tier 2 - Capacity/Archive	High-capacity HDD / object	Backups, archives, media	Best effort

12.3 RAID and Data Protection

Block storage arrays use erasure coding or RAID6/RAID-DP equivalent protection as a minimum for capacity-tier volumes, and mirrored (RAID1/RAID10 equivalent or triple-replication) protection for performance-tier volumes where write latency is the primary concern. Software-defined storage nodes use distributed erasure coding across nodes and failure domains (rack/power feed) rather than relying solely on local RAID, so that loss of an entire node or rack does not risk data loss.

12.4 Backup and Replication

Backup architecture follows a 3-2-1 model: a minimum of three copies of critical data, on two different media/storage types, with one copy stored off-site (or in a logically/physically isolated vault) to protect against ransomware and site-level events. Snapshot-based backups run against the primary storage tiers on a schedule defined per workload criticality tier, with replication to a secondary site (where a DR site exists per Section 32) using asynchronous replication for most workloads and synchronous replication for the subset of workloads requiring near-zero RPO.

Data Classification	Backup Frequency	Retention	Replication
Tier 1 - Critical production	Continuous/hourly snapshot	35 days + monthly for 1 yr	Synchronous or near-sync to DR
Tier 2 - Standard production	Daily	30 days	Asynchronous to DR
Tier 3 - Non-production	Weekly	14 days	None / best effort

13. Network Architecture

13.1 Network Topology

The data hall network fabric uses a spine-leaf (Clos) topology, replacing the traditional three-tier core-aggregation-access model to provide predictable, low-latency, non-blocking east-west connectivity between racks. Every leaf switch connects to every spine switch, so the loss of a single spine switch reduces available bandwidth but does not partition the fabric, and any leaf-to-leaf path traverses exactly one spine hop.

13.1.1 Fabric Design Parameters

Layer	Role	Phase 1 Speed	Growth Path
Spine	Fabric backbone, leaf interconnect	100GbE	400GbE
Leaf (Top-of-Rack)	Server/storage rack connectivity	25/100GbE	100/400GbE
Border/Edge leaf	WAN, internet, cloud interconnect	100GbE	400GbE
Out-of-band management	Dedicated OOB network for iDRAC/iLO/switch mgmt	1GbE	10GbE

13.2 Network Segmentation

The network is segmented into distinct zones enforced through VLANs/VXLANs and firewall policy: a production zone, a non-production/development zone, a DMZ for internet-facing services, a dedicated storage/replication network, an out-of-band management network, and a security/monitoring network carrying SIEM and DCIM telemetry. Inter-zone traffic is routed exclusively through firewalled, logged chokepoints; direct VLAN-to-VLAN routing between security zones is not permitted.

13.2.1 Segmentation Model

Zone	Purpose	Access Policy
Production	Live application and data workloads	Restricted, change-controlled
Non-production	Dev/test/staging	Isolated from production, no direct data flow
DMZ	Internet-facing services	Deny-by-default inbound, tightly filtered outbound
Storage/replication	SAN, backup, DR replication traffic	No routing to user VLANs
Out-of-band management	iDRAC/iLO, switch/PDU management	Isolated, jump-host access only
Security/monitoring	SIEM, DCIM, NOC telemetry	Read-only mirrored/tap traffic where possible

13.3 Routing and Switching Design

The fabric runs a Layer 3 (routed) design down to the leaf switch using BGP as the underlay routing protocol between leaf and spine, following the widely adopted eBGP-in-the-underlay pattern for Clos fabrics. VXLAN with EVPN control plane provides Layer 2 overlay extension where workloads (for example, virtualization live-migration domains) require Layer 2 adjacency across racks, without reintroducing a large flat Layer 2 spanning-tree domain.

13.4 Software-Defined Networking

A centralized SDN controller/fabric manager provides intent-based configuration of the leaf-spine fabric, VXLAN/EVPN overlay provisioning, and telemetry collection, reducing manual per-device configuration and configuration drift. Network changes are pushed through a version-controlled configuration pipeline (Section 26) rather than applied ad hoc to individual switches.

13.5 Load Balancing

Application load balancing is delivered through a pair of high-availability load balancer appliances (active-active or active-passive per workload requirement) positioned at the DMZ and internal application tier boundaries, providing Layer 4 and Layer 7 load balancing, SSL/TLS termination, and web application firewall (WAF) functionality for internet-facing services.

13.6 WAN and Internet Connectivity

The facility connects to the internet and to partner/cloud networks through a minimum of three physically diverse carriers, entering the building through geographically separated duct paths to eliminate a single physical cut as a total-loss event. Dedicated cloud interconnects (private, non-internet-routed circuits to major public cloud providers) are provisioned to support the hybrid-cloud workloads referenced in Section 2.1, with BGP-based failover between circuits and providers.

13.6.1 Carrier and Circuit Redundancy

- Minimum three diverse carriers, no shared physical conduit or point of entry.
- Redundant border routers in an active-active BGP configuration.
- Dedicated, redundant circuits to primary public cloud provider(s) for hybrid workloads.
- DDoS scrubbing service in-path or on-demand for internet-facing address space.

14. Cabling Infrastructure

14.1 Structured Cabling Standards

All structured cabling follows TIA-942 Rated-3 requirements, including redundant, diversely routed cable pathways between the network meet-me room, the main distribution area, and horizontal distribution areas within each data hall. Cable pathways are separated by function (power, data, out-of-band) using dedicated cable trays with defined separation distances to minimize electromagnetic interference.

14.2 Fiber and Copper Media Selection

Link Type	Media	Typical Distance	Use Case
Server/storage to leaf switch	OM4/OM5 multimode fiber or DAC	Intra-row, < 30m	25/100GbE server uplinks
Leaf to spine	OM4/OM5 or single-mode fiber	Intra-hall	100/400GbE fabric uplinks
Inter-building/inter-hall	Single-mode fiber (OS2)	Up to several km	Campus/DR interconnect
Management/OOB	Cat6A copper	< 90m	iDRAC/iLO, switch console

14.3 Cable Management and Rack Elevation Standards

Every rack is provisioned with vertical and horizontal cable management, with patch cords dressed and velcro-tied (not zip-tied, to allow non-destructive rework) and routed to maintain clear airflow paths per the containment design in Section 7.2. Rack elevation diagrams are maintained for every cabinet in the DCIM platform (Section 16) and updated as part of the mandatory change process (Section 34) before any physical work is considered complete.

14.4 Labeling and Documentation Standards

All cabling is labeled at both ends per a standardized scheme encoding data hall, row, rack, device, and port, and every physical link is registered in the DCIM/IPAM system before being placed into service. Labeling and documentation standards are mandatory, not optional, and are audited quarterly as part of the operational readiness review described in Part V.

14.4.1 Example Labeling Convention

Segment	Format	Example
Data hall	DHxx	DH01
Row	Rxx	R04
Rack	RKxx	RK12
Port/device	Device-Port	LEAF03-P24
Full label	Concatenated	DH01-R04-RK12-LEAF03-P24

15. Rack and Cabinet Design

15.1 Rack Specification

Standard cabinets are 42U-48U, 800mm wide by 1200mm deep, sized to accommodate standard-depth servers, storage arrays, and cable management side channels without obstructing hot/cold aisle airflow. High-density GPU racks use extended-depth or wider cabinets per vendor specification, with reinforced floor anchoring given the increased weight of multi-GPU chassis.

15.2 Power Density Planning

Rack power density is planned per zone (Section 6.4.1) rather than uniformly across the data hall, allowing standard-density and high-density workloads to coexist without over-provisioning cooling capacity across the entire hall. Each rack's provisioned and actual power draw is tracked in the DCIM platform against its designed capacity, with alerting when a rack trends above 80% of its provisioned circuit capacity.

15.3 Cabinet Layout and Numbering

Cabinets are numbered sequentially within each row using the labeling convention in Section 14.4.1, with row and cabinet numbering fixed at design time and never reused, even if a cabinet is decommissioned, to preserve historical documentation and audit trail integrity.

16. Data Center Infrastructure Management (DCIM)

16.1 DCIM Platform Overview

A centralized DCIM platform provides the single source of truth for physical asset inventory, rack elevation, power and cooling capacity, cable connectivity, and environmental monitoring across the facility. The DCIM platform integrates with the building management system (BMS) for power and cooling telemetry, with the network management system for switch/port inventory, and with the IT service management (ITSM) platform for asset and change records.

16.2 Core DCIM Functions

- Real-time power monitoring at the utility, UPS, PDU, and rack circuit level, feeding the PUE/WUE calculations in Section 7.4.
- Environmental monitoring: temperature and humidity sensors at multiple points per row, feeding containment and cooling alarms.
- Asset and rack elevation management, including make/model/serial and warranty tracking for every installed device.
- Capacity planning: available power, cooling, space, and network port capacity by data hall, row, and rack.
- Cable and connectivity (IPAM/DCIM-integrated) records for every physical link, per Section 14.4.

16.3 Integration with Monitoring and Ticketing

DCIM alarms (power threshold breaches, environmental excursions, capacity threshold breaches) are integrated with the NOC monitoring stack described in Part V, Section 29, and automatically generate incident or problem tickets in the ITSM platform per the thresholds defined in the operations runbooks.

17. Cybersecurity Architecture

17.1 Network Security Controls

Perimeter and inter-zone network security is enforced through next-generation firewalls providing stateful inspection, intrusion prevention (IPS), and application-layer visibility at every segmentation boundary defined in Section 13.2. All firewall rule changes follow the change management process in Section 26, with quarterly rule-base reviews to remove stale or overly permissive rules.

17.2 Zero Trust Architecture

Beyond perimeter and network segmentation controls, the design implements Zero Trust principles for east-west access: no implicit trust is granted based on network location alone. Workload-to-workload communication is authenticated and authorized at the application/identity layer (e.g., mutual TLS, service mesh policy) in addition to network-layer segmentation, and administrative access to infrastructure is brokered through a privileged access management (PAM) solution rather than direct credentialed login.

17.3 Data Encryption

Data at rest on storage tiers holding sensitive or regulated data is encrypted using array-level or software-defined-storage-level encryption with keys managed through a centralized key management system (KMS), separate from the storage platform itself. Data in transit between data halls, to the DR site, and across WAN/cloud interconnects is encrypted using IPsec or MACsec as appropriate to the link type, in addition to any application-layer TLS.

17.4 Identity and Access Management

All human and service access to infrastructure is provisioned through a centralized identity provider with role-based access control (RBAC), multi-factor authentication for all administrative access, and just-in-time privilege elevation for high-risk operations rather than standing administrative rights. Access reviews are conducted quarterly, with automated de-provisioning triggered by HR/identity lifecycle events.

17.4.1 Security Monitoring

A Security Information and Event Management (SIEM) platform aggregates logs from network devices, servers, hypervisors, identity systems, and physical security systems (badge access, mantrap events) described in Section 9, providing correlated detection and a 24/7-monitored alerting pipeline into the security operations function.

17A. IP Addressing, DNS, DHCP, and Time Services

17A.1 IP Address Management (IPAM)

IP address allocation is managed centrally through an IPAM system integrated with the DCIM/CMDB platform (Section 16), assigning address space per network zone (Section 13.2) from a structured, hierarchical addressing plan that leaves growth headroom within each zone for the full Phase 1-4 build-out without requiring re-addressing of live systems.

17A.1.1 Representative Addressing Plan

Zone	Address Block Purpose	Allocation Approach
Production compute	Server/VM host addressing	Per-rack or per-row subnet, summarized at leaf
Production storage	SAN/NAS/replication traffic	Dedicated non-routed block, isolated from user VLANs
Container platform	Kubernetes pod/service networking	Dedicated overlay CIDR per cluster, non-overlapping
Out-of-band management	iDRAC/iLO, switch/PDU management	Dedicated block, firewalled, jump-host only access
DMZ	Internet-facing services	Public/NAT'd block, tightly filtered

17A.2 DNS Architecture

Internal DNS is served by a minimum of two geographically/logically separated DNS servers per zone, with authoritative zones for internal infrastructure names replicated across both, and conditional forwarding configured for hybrid-cloud name resolution. External-facing DNS for internet services uses a managed, DDoS-resilient authoritative DNS provider with secondary DNS redundancy.

17A.3 DHCP and Network Boot Services

DHCP services support both standard client addressing (where applicable) and PXE network-boot for bare-metal server provisioning, integrated with the server lifecycle automation described in Section 17C. DHCP scopes are aligned with the IPAM plan and are deployed in a failover/redundant configuration to avoid a single DHCP server outage impacting provisioning workflows.

17A.4 Time Synchronization

All infrastructure devices synchronize time via NTP from a redundant, internally hosted stratum-2 (or better) time source, itself synchronized from multiple external stratum-1 references. Accurate, consistent time synchronization is a prerequisite for reliable log correlation in the SIEM platform (Section 17.4.1) and for certain authentication protocols used in the identity architecture.

17B. Network Device Standards and Configuration Management

17B.1 Switch and Router Platform Standards

Leaf, spine, and border network devices are standardized on a small number of qualified platforms per role to simplify spares holding, firmware management, and operational training, consistent with the server standardization approach in Section 11.2. Platform selection prioritizes support for the EVPN/VXLAN overlay architecture (Section 13.3), open, programmatic management interfaces, and vendor support SLAs aligned to the targets in Section 35.2.

17B.2 Configuration Management and Infrastructure as Code

Network device configuration is managed through an infrastructure-as-code pipeline: desired-state configuration is defined in version-controlled templates, validated in a pre-production/lab environment, and pushed to production devices through the SDN controller/automation pipeline referenced in Section 13.4, rather than through direct manual CLI configuration of production devices.

17B.2.1 Configuration Change Workflow

- Change proposed as a pull request against the version-controlled configuration repository.
- Automated syntax and policy validation (e.g., no overly permissive ACLs, no VLAN overlap) run against the proposed change.
- Peer review and approval per the change management thresholds in Section 26.1/34.1.
- Staged rollout: lab/non-production first, then a canary subset of production devices, then full rollout.
- Automated rollback triggered if post-change health checks fail.

17B.3 Firmware and Patch Management

Network device, server, and storage firmware is managed on a defined update cadence, balancing security patching urgency against change-risk: critical security firmware advisories are assessed within 5 business days and scheduled for the next available maintenance window per the MOP risk framework in Section 28.3, while routine firmware updates follow a quarterly cadence tested first in the non-production environment.

17C. Server and Infrastructure Lifecycle Automation

17C.1 Bare-Metal Provisioning

New server hardware is provisioned through an automated bare-metal provisioning pipeline: PXE boot triggers an automated OS/hypervisor installation, applies the standardized firmware and BIOS configuration baseline, registers the asset in the DCIM/CMDB, and hands the host off to the configuration management/orchestration layer for role assignment, eliminating manual, inconsistent server build processes.

17C.2 Configuration Baseline and Drift Detection

All servers and hypervisor hosts are built against a version-controlled configuration baseline (OS hardening settings, monitoring agent installation, logging configuration, time sync, patch agent). Configuration drift from this baseline is detected continuously through the configuration management tooling and either auto-remediated or flagged for review, depending on the nature of the drift.

17C.3 Patch Management Program

Patch Category	Assessment SLA	Deployment Target
Critical/actively exploited vulnerability	24 hours	Emergency change, within 72 hours
High severity	5 business days	Next scheduled maintenance window (≤ 30 days)
Medium/Low severity	Monthly review	Quarterly patch cycle

17C.4 Decommissioning and Secure Data Erasure

Hardware decommissioning follows a documented process: workload evacuation and validation, removal from monitoring/DCIM/CMDB active inventory (retained in historical records), secure data erasure or physical destruction of storage media per the data classification handling requirements, and removal from the physical rack with corresponding DCIM elevation and cabling record updates (Section 14.4).

17D. Additional Cybersecurity Program Elements

17D.1 Vulnerability Management

Infrastructure is scanned for vulnerabilities on a continuous/near-continuous basis using authenticated scanning where possible, with findings triaged against the patch management SLAs in Section 17C.3 and tracked to closure in the same problem-management backlog used for operational issues (Section 31.3).

17D.2 Penetration Testing and Red Team Exercises

Independent third-party penetration testing is conducted at least annually against the network perimeter, internal segmentation controls, and physical security controls, with findings remediated per severity-based SLAs and validated through re-testing. Periodic red-team exercises assess detection and response capability of the SOC/NOC function described in Section 17.4.1 and Part V.

17D.3 Security Operations Center (SOC) Function

The SIEM-driven security monitoring function (Section 17.4.1) operates as a 24/7 Security Operations Center capability, either staffed on-site as part of the NOC organization (Section 27) or delivered through a managed security service provider under a defined SLA, responsible for triage and initial response to security alerts before escalation to the incident management process (Section 31).

17D.4 Compliance Auditing

Where the facility hosts workloads subject to regulatory compliance frameworks, periodic internal and external audits validate control effectiveness against the applicable framework, with audit findings tracked through the same change and problem management processes used for operational and security findings, ensuring a single system of record for corrective actions.

17D.5 Supply Chain and Hardware Security

Hardware and firmware sourced for the facility follows a defined supply-chain risk management process, including sourcing only from vendors on the approved vendor list (Section 22.2), firmware integrity verification prior to deployment, and hardware-root-of-trust/secure-boot features enabled where supported, to reduce the risk of compromised hardware or firmware entering the environment.

17E. Storage Protocol and Data Services Detail

17E.1 Protocol Selection Guidance

Protocol	Transport	Typical Use Case
Fibre Channel	Dedicated FC fabric	Latency-sensitive block storage, legacy enterprise arrays
NVMe-oF (TCP or RoCE)	Ethernet fabric	High-performance block storage over converged network
iSCSI	Ethernet fabric	General-purpose block storage, lower cost hosts
NFS/SMB	Ethernet fabric	File shares, VM datastores, general file services
S3-compatible object	Ethernet fabric (HTTP/HTTPS)	Backup targets, unstructured data, application object storage

17E.2 Data Reduction and Efficiency

Storage platforms apply inline deduplication and compression where workload characteristics support it (particularly on virtualization datastores and backup targets), with efficiency ratios monitored to ensure raw-to-usable capacity planning assumptions (feeding Section 33) remain accurate as the workload mix evolves.

17E.3 Storage Quality of Service

Performance-tier storage implements QoS policies to prevent a single noisy-neighbor workload from degrading latency for co-located tenants on shared storage infrastructure, with minimum/maximum IOPS and bandwidth limits assignable per volume or per tenant.

17F. Network Resilience Testing and Validation

17F.1 Fabric Failure Testing

The spine-leaf fabric (Section 13.1) is validated during commissioning (Section 20.3) through controlled failure injection: individual spine switches, leaf switches, and links are taken out of service in turn to confirm traffic reconverges within the design threshold and no single failure isolates a rack from the fabric.

17F.2 Ongoing Resilience Validation

Beyond initial commissioning, network resilience is periodically re-validated through scheduled chaos-engineering style exercises in the non-production environment and, for select low-risk scenarios, in production during defined maintenance windows, to catch configuration drift or capacity growth that could have silently eroded the fabric's designed failure tolerance.

17G. Representative Network Hardware Bill of Materials (Phase 1)

The following table illustrates a representative Phase 1 network hardware bill of materials consistent with the architecture in Sections 13-15. Exact model numbers are subject to the vendor selection process in Section 22.2 and are intentionally omitted; port counts and speeds reflect the design intent.

Item	Role	Quantity (Phase 1)	Port Configuration
Spine switch	Fabric backbone	4	32x 100GbE (growth to 400GbE)
Leaf / Top-of-Rack switch	Rack server/storage connectivity	24 (2 per rack, A/B)	48x 25GbE + 6x 100GbE uplink
Border/edge router	WAN, internet, cloud interconnect	2 (redundant pair)	8x 100GbE
Out-of-band switch	iDRAC/iLO/switch management	12	48x 1GbE
Load balancer appliance	L4/L7 load balancing, WAF	2 (HA pair)	4x 25GbE
Next-generation firewall	Perimeter and inter-zone segmentation	2 (HA pair per zone boundary)	8x 25/100GbE

17G.1 Cabling Bill of Materials (Representative, Phase 1)

Cable Type	Approx. Quantity (Phase 1)	Primary Use
OM4 MPO/LC fiber trunk (24-strand)	120 runs	Leaf-to-spine, row-to-MDA backbone
Single-mode fiber (OS2)	40 runs	Inter-hall, carrier entrance, DR link
DAC/AOC 25GbE	480 units	Server-to-leaf short-range links
Cat6A copper	300 runs	Out-of-band management, console access

17H. Infrastructure Automation and Orchestration Stack

17H.1 Infrastructure-as-Code Layer

Compute, storage, and network resources are provisioned and managed through an infrastructure-as-code toolchain, with declarative configuration held in version control and applied through an automated pipeline rather than manual console or CLI changes. This applies uniformly to the physical provisioning automation described in Section 17C, the network configuration pipeline described in Section 17B.2, and to virtualization/container platform resource definitions.

17H.2 Orchestration and Scheduling

The virtualization layer (Section 11.3) and container platform (Section 11.4) each run their native orchestration/scheduling engine, with placement policy (anti-affinity across racks/power feeds, resource reservations for latency-sensitive workloads) enforced automatically rather than relying on manual placement decisions, ensuring the redundancy assumptions in Section 10.1 are maintained as workloads are scheduled and rescheduled over time.

17H.3 CI/CD Integration for Infrastructure Changes

Infrastructure-as-code changes flow through the same staged validation pipeline described in Section 17B.2.1: automated linting and policy checks, peer review, staged rollout to non-production, canary rollout to a subset of production, then full rollout, with automated rollback triggers based on post-change health checks integrated with the monitoring stack (Section 29).

17I. Observability Stack

17I.1 Metrics, Logs, and Traces

Observability is built on three complementary data types: metrics (time-series numeric data such as CPU utilization or link throughput, feeding the dashboards in Section 29.3), logs (structured and unstructured event records from every infrastructure component, aggregated centrally and feeding the SIEM described in Section 17.4.1), and distributed traces (for containerized/microservice workloads, allowing a single request to be followed across multiple services to identify latency contributors).

17I.2 Retention and Storage Tiering for Telemetry

Data Type	Hot Retention (fast query)	Cold Retention (archive)
Infrastructure metrics	13 months	None beyond hot tier (per FR-008, Section 4A.1)
Security/audit logs	90 days	1-7 years depending on data classification
Application/system logs	30-90 days	1 year (compliance-driven workloads)
Distributed traces	7-14 days	None (sampled, high-cardinality data)

17I.3 Dashboards and Correlation

A unified observability platform correlates metrics, logs, and traces against the same asset identifiers used in the DCIM/CMDB (Section 16), so an engineer investigating a performance issue can pivot directly from an application trace to the underlying host, rack, and power/cooling telemetry for that physical location.

17J. Database and Data Platform Architecture Guidance

17J.1 Database Deployment Patterns

Relational databases supporting Tier 1 transactional workloads are deployed in a clustered, synchronously-replicated configuration across the A/B power and network fabric domains described in Section 10.1, so that loss of a single rack's power or network path does not require a database failover. Read-heavy workloads are offloaded to asynchronously replicated read replicas to reduce load on the primary write path.

17J.2 Storage Placement for Database Workloads

Database data files are placed on the Tier 0 performance storage tier (Section 12.2.1) given their latency sensitivity, while backup and archive data for the same databases is placed on Tier 2 capacity storage per the backup architecture in Section 12.4, keeping the expensive performance tier reserved for active working sets.

17J.3 Analytics and Data Platform

Analytics and machine learning data platforms are deployed on dedicated node pools (Section 11.4.1) separate from transactional database infrastructure, with data ingested from operational systems through defined, access-controlled pipelines rather than direct query access to production transactional databases, both for performance isolation and for data governance.

17K. High-Density and Liquid Cooling Readiness

17K.1 Air-Cooling Density Limits

The Phase 1 air-cooling design (Section 7.1) supports rack densities up to approximately 30 kW using in-row or rear-door heat exchanger units; beyond this density, air-cooling alone becomes impractical due to the airflow volume required, and the design incorporates a defined transition path to direct liquid cooling for future GPU generations.

17K.2 Direct Liquid Cooling Design Provision

High-density GPU rows (Section 15.1) are provisioned at Phase 1 construction with the facility-side piping stubs, floor loading, and mechanical plant headroom needed to support a future direct-to-chip liquid cooling loop (a technology cooling loop isolated from the facility chilled-water loop via heat exchangers), avoiding the need for disruptive retrofit of gray-space piping when liquid-cooled hardware is deployed in a later phase.

17K.3 Liquid Cooling Safety and Leak Mitigation

Where direct liquid cooling is deployed, rack-level leak detection, automatic isolation valves, and drip trays are required in addition to the facility-level leak detection described in Section 9B.4, and MOPs for liquid-cooled rack maintenance (Section 28.1) include specific pre-work leak-check and coolant-handling steps beyond the standard electrical/mechanical MOP framework.

17L. Service Mesh and API Gateway Architecture

17L.1 Service-to-Service Communication

Within the container platform (Section 11.4), a service mesh provides mutual TLS encryption, fine-grained traffic policy, and telemetry for service-to-service communication, implementing the workload-to-workload Zero Trust controls described in Section 17.2 without requiring every application team to individually implement encryption and authentication logic.

17L.2 API Gateway

External and cross-domain API traffic is routed through a centralized API gateway providing authentication, rate limiting, and request/response logging, positioned at the DMZ boundary described in Section 13.2 and integrated with the load balancing and WAF layer described in Section 13.5.

17M. Wireless and Mobile Connectivity Within the Facility

17M.1 Wi-Fi for Operations and Maintenance

A dedicated operations Wi-Fi network, isolated from all production data VLANs, supports handheld scanning devices, mobile DCIM applications used during rounds (Section 27A.2), and technician tablets used to reference MOPs on the data hall floor. This network is treated as an untrusted segment from the perspective of the production network and is subject to the same firewalled-chokepoint policy described in Section 13.2.

17M.2 Distributed Antenna System / Cellular Coverage

Where the building envelope attenuates cellular signal, a distributed antenna system (DAS) or cellular repeater ensures staff can reliably receive emergency notifications and communicate during an incident from anywhere in the facility, including gray-space plant rooms.

17N. Asset and Software License Management

17N.1 Hardware Asset Lifecycle Tracking

Every physical asset is tracked from receipt (goods-in inspection against purchase order) through installation (Section 17C.1), operational life (linked to maintenance history, Section 30.3), and decommissioning (Section 17C.4) in the DCIM/CMDB, providing full lifecycle traceability used both for warranty claims and for capital planning refresh cycles (Section 11.2).

17N.2 Software License Management

Software licensing for hypervisor, container platform, database, backup, DCIM, and monitoring tooling is tracked centrally against entitlement records, with utilization reconciled quarterly against purchased entitlements to avoid both compliance exposure and unnecessary over-purchasing ahead of demonstrated need.

17O. Network Time and Certificate Management

17O.1 Internal Public Key Infrastructure

An internal certificate authority issues and manages certificates for internal service-to-service TLS (feeding the service mesh described in Section 17L.1), device management interfaces, and internal web applications, with automated certificate issuance and renewal to avoid outages caused by expired certificates, and a defined root/intermediate CA key protection process consistent with the KMS practices described in Section 17.3.

17O.2 Certificate Lifecycle Automation

Certificate issuance and renewal are automated wherever the consuming platform supports it (container platform ingress, service mesh, internal load balancers), with manual certificate management reserved only for systems that cannot support automated renewal, each of which is tracked in a dedicated inventory with expiry alerting well ahead of the renewal deadline.

17P. Multi-Tenancy and Workload Isolation Model

17P.1 Tenancy Model

Where the facility hosts workloads on behalf of multiple internal business units or, in a colocation-adjacent model, external tenants, isolation is enforced at network (Section 13.2), compute (dedicated resource pools or hard multi-tenancy per Section 11.3.1), and storage (Section 17E.3 QoS and, where required, dedicated storage pools) layers, with the isolation model documented per tenant in the CMDB.

17P.2 Tenant Onboarding Process

New tenant onboarding follows a defined process: network zone and address space allocation (Section 17A.1), resource quota provisioning on the virtualization/container platform, access provisioning through the identity system (Section 17.4), and a security review confirming the tenant's isolation requirements are met before production workloads are migrated.

17Q. Edge and CDN Integration

For workloads with an external content-delivery or edge-caching requirement, the facility's border network (Section 13.6) integrates with content delivery network (CDN) providers via the same dedicated interconnect model used for cloud connectivity, with origin-shielding and cache-control policy defined at the application layer to minimize unnecessary origin traffic back to the facility.

17R. Data Lifecycle and Retention Governance

17R.1 Data Retention Policy Alignment

Storage and backup retention settings (Section 12.4) are configured per the data classification and retention schedule defined in the organization's records management policy, with automated lifecycle policies (tiering to lower-cost storage, expiry/deletion) applied wherever the platform supports it to avoid manual, error-prone retention management.

17R.2 Secure Deletion and Right-to-Erasure Handling

Where applicable regulatory frameworks require it, the data platform supports identifiable deletion of a specific data subject's records across primary, backup, and archive tiers within the regulatory timeframe, a capability validated as part of the compliance audit process described in Section 17D.4.

17S. Capacity Modeling Detail for IT Infrastructure

17S.1 Compute Capacity Model

Metric	Phase 1 Design Target
vCPU-to-physical-core ratio (general workloads)	4:1 to 6:1 depending on workload profile
Memory overcommit ratio	1.2:1 maximum for production workloads
Target average cluster utilization	60-70% (headroom for N+1 host maintenance, Section 11.3.1)
GPU allocation model	Dedicated (no oversubscription) for training; shared/time-sliced permitted for inferencing

17S.2 Storage Capacity Model

Metric	Phase 1 Design Target
Tier 0 performance headroom	30% free capacity maintained at all times
Tier 1 general purpose headroom	25% free capacity maintained at all times
Tier 2 capacity/archive headroom	20% free capacity maintained at all times
Data reduction assumption (dedupe/compression)	1.5:1 to 3:1 depending on workload, validated quarterly (Section 17E.2)

17S.3 Network Capacity Model

Leaf-to-spine oversubscription ratio is designed to 3:1 at Phase 1, reviewed against actual east-west traffic patterns captured through the observability stack (Section 17I) and tightened toward 1:1 in specific high-throughput rows (e.g., GPU training clusters using RDMA, Section 11.2) where workload characteristics justify the additional spine capacity.

17T. Full Reference Server BIOS/Firmware Baseline

Setting	Standard Value	Rationale
Power profile	OS-controlled / performance	Consistent power behavior under hypervisor control
Hyper-threading / SMT	Enabled (workload-dependent exceptions documented)	Maximizes core throughput for general workloads
NUMA configuration	Enabled, node interleaving disabled	Preserves memory locality for latency-sensitive workloads
Boot mode	UEFI with Secure Boot enabled	Supply chain / firmware integrity (Section 17D.5)
TPM	Enabled	Supports disk encryption key protection
Out-of-band management (iDRAC/iLO)	Enabled, isolated VLAN only (Section 17A.1)	Lights-out management without exposure on production network
Watchdog timer	Enabled	Automatic recovery from hung POST/boot state

17U. Representative Network ACL and Firewall Policy Baseline

Zone Pair	Default Policy	Notes
DMZ → Production	Deny by default, explicit allow per approved flow	Section 13.2 chokepoint enforcement
Production → DMZ	Deny by default, explicit allow per approved flow	Prevents lateral pivot from production outward
Non-production → Production	Deny	Hard isolation per Section 13.2
Out-of-band mgmt → Any	Deny except jump-host initiated	Section 17.4 PAM enforcement
Storage/replication → User VLANs	Deny	Section 13.2 segmentation model
Any → Security/monitoring	Allow (mirrored/tap only, one-way)	Read-only telemetry collection

17V. Detailed Container Platform Resource Quota Model

Namespace Tier	CPU Request/Limit Ratio	Memory Request/Limit Ratio	Priority Class
System/critical add-ons	1:1	1:1	Highest - never preempted
Production application	1:2	1:1.5	High
Batch/analytics	1:4	1:2	Medium - preemptible
Development/sandbox	1:1 (hard cap)	1:1 (hard cap)	Low - first preempted

17W. Detailed Storage Snapshot and Retention Policy Matrix

Workload	Snapshot Frequency	Local Retention	Remote Copy Frequency	Remote Retention
Tier 1 OLTP database	Every 15 minutes	48 hours	Continuous (near-sync)	35 days
Tier 1 application VM	Hourly	7 days	Every 4 hours	35 days
Tier 2 general VM	Daily	14 days	Daily	30 days
Tier 2 file share	Daily	14 days	Daily	30 days
Tier 3 non-production	Weekly	14 days	None	N/A
Object/archive store	N/A (immutable write-once)	Per retention policy (Section 17R.1)	Cross-region replication	Per retention policy

17X. Network Cable and Fiber Test Acceptance Criteria

Test	Acceptance Criteria	Standard Reference
Copper Cat6A permanent link	Pass per TIA-568 Category 6A limits	TIA-568
Multimode fiber (OM4/OM5) insertion loss	≤ 0.35 dB per connector, ≤ 1.5 dB per link (typical)	TIA-568 / TIA-942
Single-mode fiber insertion loss	≤ 0.5 dB per connector (typical)	TIA-568
OTDR trace	No unexplained events > 0.3 dB, no fiber breaks	Section 30.2 annual audit
Polarity verification	Correct transmit/receive mapping end to end	TIA-568

17Y. Sample DCIM Data Model - Key Object Types

Object Type	Key Attributes
Data Hall	Name, floor area, design power/cooling capacity, current utilization
Row / Rack	Location code (Section 14.4.1), power circuit assignment, U-capacity, weight capacity
Device (server/switch/storage)	Asset tag, make/model/serial, rack position (U), power feed(s), warranty end date
Port / Connection	Device, port ID, connected-to device/port, cable type and length, VLAN/label
Circuit	Source panel, destination rack/PDU, rated capacity, current load, redundancy pair
Sensor	Type (temp/humidity/leak), location, current reading, alarm thresholds

17Z. Application Tiering and Placement Policy

Application Tier	Placement Rule	Redundancy Expectation
Tier 1 - Critical	Anti-affinity across racks and power feeds mandatory	N+1 minimum at every layer (compute, network, storage)
Tier 2 - Standard	Anti-affinity across racks recommended	N+1 at cluster level; single-rack impact tolerable briefly
Tier 3 - Non-production	No placement constraint	Best effort, no redundancy guarantee

18. Project Governance and Organization

18.1 Governance Structure

The Project is governed by a Steering Committee comprising executive sponsors, the Project Director, and lead representatives from Facilities, IT Infrastructure, Network Engineering, Security, and Finance. The Steering Committee meets monthly to review schedule, budget, and risk status, and holds authority over scope changes exceeding the thresholds defined in Section 26.

18.2 Project Organization

Role	Responsibility
Project Director	Overall delivery accountability, Steering Committee liaison
Facility Design Lead	Civil, electrical, mechanical, fire, physical security design
IT/Network Architecture Lead	Compute, storage, network, cabling design (Part III)
Construction Manager	General contractor and trade coordination, site safety
Commissioning Manager	Level 1-5 commissioning program (Section 20.3)
Security Lead	Physical and cyber security design and implementation
Operations Readiness Lead	Staffing, SOPs, transition to operations (Part V)
PMO / Project Controls	Schedule, budget, risk, and change tracking

18.3 Decision Rights and Escalation

Day-to-day design and delivery decisions are made by the relevant discipline lead within their approved budget and schedule envelope. Decisions affecting Tier rating, overall budget, or the committed schedule milestones require Project Director approval, with changes above the thresholds in Section 26 escalated to the Steering Committee.

19. Work Breakdown Structure

The Project is decomposed into six top-level work packages aligned to the document structure: (1) Site and Facility Shell, (2) Electrical Infrastructure, (3) Mechanical Infrastructure, (4) Fire and Security Systems, (5) IT and Network Infrastructure, and (6) Commissioning and Transition to Operations. Each work package is further decomposed into a detailed WBS maintained in the project scheduling tool, summarized at a high level below.

WBS Item	Key Deliverables
1. Site & Facility Shell	Site prep, civil works, structural shell, raised floor
2. Electrical Infrastructure	Utility interconnect, switchgear, UPS, generators, PDUs
3. Mechanical Infrastructure	Chiller plant, CRAH/in-row units, containment, BMS
4. Fire & Security Systems	VESDA, suppression, access control, CCTV, mantrap
5. IT & Network Infrastructure	Racks, cabling, network fabric, compute/storage Phase 1
6. Commissioning & Transition	Level 1-5 commissioning, SOP development, staffing, go-live

20. Project Schedule and Milestones

20.1 Overall Schedule

The Project follows an 18-month schedule from design freeze to commissioning completion, structured into five stages: Design Development (months 1-3), Permitting and Procurement (months 2-5, overlapping design), Construction (months 4-14), IT Infrastructure Fit-Out (months 11-16, overlapping late construction), and Commissioning and Transition to Operations (months 15-18).

20.2 Milestone Schedule

Milestone	Target Month
Design freeze and permitting submission	M3
Long-lead equipment orders placed (generators, chillers, switchgear)	M3
Site mobilization and civil works start	M4
Building shell weathertight	M9
Electrical infrastructure energized (utility power live)	M12
Mechanical plant operational	M12
IT infrastructure fit-out (Phase 1 racks, network, compute)	M14
Level 1-4 commissioning complete	M16
Level 5 integrated systems test complete	M17
Operational readiness review and go-live	M18

20.3 Commissioning Levels

Commissioning follows the standard five-level model used in data center delivery: Level 1 (factory/component testing), Level 2 (site acceptance testing of individual components), Level 3 (system-level functional testing, e.g., UPS system as a whole), Level 4 (integrated systems testing across electrical and mechanical systems together), and Level 5 (full integrated systems test under simulated load and failure scenarios, including utility failure and generator start, UPS transfer, and cooling failover).

20.4 Phased Capacity Triggers

Phases 2-4 (Section 1.1) are not calendar-driven; each phase is triggered when sustained utilization of the prior phase's critical IT load crosses 75% for two consecutive months, giving the PMO a lead-time trigger to initiate long-lead procurement for the next phase's electrical and mechanical capacity before the prior phase is exhausted.

21. Budget and Cost Management

21.1 Cost Management Approach

The Project budget is managed against the cost categories defined in Section 3.3, with monthly cost-to-complete forecasting against the approved baseline. A management contingency reserve equal to 10% of total Phase 1 CapEx is held at the Project Director level and released only through formal change control (Section 26); a separate 5% design/construction contingency is held within each discipline's budget for routine field variances.

21.2 Cost Reporting

Monthly cost reports track committed cost, actual cost, and forecast-at-completion against the baseline budget by WBS category, with variance explanations required for any category tracking more than 5% over baseline. Cost reports are reviewed by the PMO and presented to the Steering Committee monthly alongside schedule status.

21.3 Earned Value Management

Schedule and cost performance are tracked using earned value management (EVM), with Schedule Performance Index (SPI) and Cost Performance Index (CPI) calculated monthly at the work-package level. An SPI or CPI below 0.9 for two consecutive months triggers a mandatory recovery plan from the responsible discipline lead.

22. Procurement and Vendor Management

22.1 Procurement Strategy

Long-lead equipment (generators, chillers, medium-voltage switchgear, UPS systems) is procured under a fixed-price, fixed-schedule contract structure to protect the Project from cost escalation and schedule risk on the critical path. IT and network hardware procurement is structured to allow late-stage vendor/model selection where possible, so Phase 1 fit-out uses current-generation hardware rather than specifications frozen at design start.

22.2 Vendor Selection Criteria

Criterion	Weight
Technical compliance with specification	30%
Total cost (capital + support/maintenance)	25%
Track record / references at comparable scale	20%
Support model and local service presence	15%
Financial stability of vendor	10%

22.3 Contract and SLA Structure

Major equipment vendors and the general contractor operate under contracts including defined performance guarantees (e.g., generator load-bank test results, UPS efficiency figures, PUE commissioning targets) with liquidated damages provisions for schedule slippage on critical-path deliverables. Ongoing maintenance contracts (Part V, Section 30) are negotiated separately with response-time SLAs tiered by criticality.

23. Risk Management Plan

23.1 Risk Management Process

Project risks are logged in a risk register maintained by the PMO, scored on a standard 5x5 likelihood/impact matrix, and reviewed at each Steering Committee meeting. Risks scoring in the top quadrant (high likelihood, high impact) require a documented mitigation owner and target closure date.

23.2 Key Project Delivery Risks

Risk	Likelihood	Impact	Mitigation Owner	Mitigation
Long-lead equipment delivery delay	Medium	High	Procurement Lead	Fixed-schedule contracts, early order placement
Labor/trade availability constraints	Medium	Medium	Construction Manager	Early trade contracting, schedule float
Design changes during construction	Medium	Medium	Facility Design Lead	Design freeze gate at M3, formal change control
Commissioning defects requiring rework	Medium	High	Commissioning Manager	Staged commissioning (Level 1-5), early defect capture
IT hardware supply chain delay	Medium	Medium	IT/Network Architecture Lead	Late-stage procurement, multiple qualified vendors

23.3 Contingency Planning

Each top-quadrant risk carries a documented contingency response (e.g., alternate vendor pre-qualified, schedule float allocated, temporary generator rental arranged) reviewed and refreshed at each monthly Steering Committee meeting through Project close.

24. Quality Management Plan

24.1 Quality Standards

All design and construction work is delivered against the standards catalogue in Appendix B, with discipline leads responsible for confirming design compliance before each stage gate. Quality is verified through a combination of third-party inspection (structural, electrical, mechanical), factory witness testing for major equipment, and the staged commissioning program in Section 20.3.

24.2 Quality Assurance Activities

- Design reviews at 30%, 60%, 90%, and 100% design completion, with discipline and cross-discipline (clash detection) review.
- Factory acceptance testing (FAT) witnessed by Project engineers for UPS, generators, and switchgear before shipment.
- Site acceptance testing (SAT) for all major equipment upon installation, prior to energization.
- Independent commissioning agent oversight of Level 4-5 integrated systems testing.

24.3 Punch List and Defect Management

Defects identified during commissioning are logged in a punch list tracked to closure before the corresponding system is accepted into operations, with critical-path defects (those blocking a milestone in Section 20.2) escalated daily until resolved.

25. Communication Plan

25.1 Reporting Cadence

Report/Meeting	Frequency	Audience
Steering Committee review	Monthly	Executive sponsors, discipline leads
PMO schedule/cost report	Monthly	Steering Committee, discipline leads
Discipline lead coordination meeting	Weekly	Discipline leads, PMO
Site safety and progress meeting	Weekly	Construction Manager, trade contractors
Commissioning status review	Weekly during commissioning stage	Commissioning Manager, discipline leads
Risk register review	Monthly	PMO, discipline leads

25.2 Stakeholder Communication

Communication to stakeholders outside the core project team (finance, business unit sponsors, future operations staff) follows a monthly summary status report, with dedicated briefings ahead of each major milestone in Section 20.2 and a formal readiness briefing ahead of the go-live decision.

26. Change Management

26.1 Change Control Process

All changes to approved scope, design, schedule, or budget are submitted through a formal change request, assessed for impact by the relevant discipline lead and the PMO, and approved at the level appropriate to the change's cost and schedule impact per the thresholds below.

Change Impact	Approval Authority
< \$50,000 and no schedule impact	Discipline Lead
\$50,000 - \$250,000 or minor schedule impact	Project Director
> \$250,000 or critical-path schedule impact	Steering Committee
Any change affecting Tier rating or life-safety systems	Steering Committee + independent commissioning agent review

26.2 Configuration and Design Baseline Control

Once design freeze (Section 20.2, M3) is reached, all design documents are placed under formal baseline control; subsequent changes are tracked as deviations from baseline with full traceability, feeding both the as-built documentation set and the DCIM platform system-of-record described in Section 16.

20A. Detailed Schedule by Work Package

20A.1 Site and Facility Shell Schedule

Task	Start	Duration	Predecessor
Site survey and geotechnical investigation	M1	4 weeks	-
Site grading and utility trenching	M4	8 weeks	Design freeze
Foundation and slab	M5	6 weeks	Grading complete
Structural shell erection	M6	10 weeks	Foundation complete
Building envelope (roof, weatherproofing)	M8	6 weeks	Shell erected
Raised floor / interior fit-out shell	M9	8 weeks	Envelope complete

20A.2 Electrical Infrastructure Schedule

Task	Start	Duration	Predecessor
Utility interconnect application and design review	M1	12 weeks	-
Long-lead switchgear/UPS/generator order placement	M3	-	Design freeze
Medium-voltage switchgear installation	M9	6 weeks	Shell weathertight
Generator plant installation and fuel system	M9	8 weeks	Shell weathertight
UPS installation	M10	6 weeks	Switchgear installed
PDU/RPP installation (white space)	M11	6 weeks	UPS installed
Electrical energization and Level 2/3 testing	M12	4 weeks	All electrical installed

20A.3 Mechanical Infrastructure Schedule

Task	Start	Duration	Predecessor
Chiller plant order and delivery	M3	-	Design freeze (long-lead)
Mechanical yard equipment installation	M9	6 weeks	Shell weathertight

Task	Start	Duration	Predecessor
CRAH / in-row unit installation (white space)	M10	6 weeks	Raised floor complete
Containment installation	M11	3 weeks	CRAH units installed
BMS integration and controls commissioning	M11	4 weeks	Mechanical equipment installed
Mechanical Level 2/3 testing	M12	4 weeks	All mechanical installed

20A.4 IT and Network Fit-Out Schedule

Task	Start	Duration	Predecessor
Rack and cable pathway installation	M11	4 weeks	Raised floor / containment ready
Structured cabling installation and certification	M12	5 weeks	Racks installed
Network fabric hardware installation and cabling	M13	4 weeks	Structured cabling certified
Compute/storage hardware installation (Phase 1)	M14	4 weeks	Network fabric operational
DCIM/monitoring stack deployment and integration	M14	3 weeks	Overlaps hardware installation
IT Level 3/4 functional testing	M15	3 weeks	Hardware installed and configured

21A. Resource and Staffing Plan (Delivery Phase)

21A.1 Project Delivery Team Resourcing

Role	Phase 1 FTE Equivalent	Engagement Window
Project Director	1.0	Full project duration
Facility Design Lead + design team	3-5	Design through construction support
IT/Network Architecture Lead + team	3-4	Design through IT fit-out and commissioning
Construction Manager + site team	2-3 (plus trade subcontractors)	Mobilization through substantial completion
Commissioning Manager + agents	2-3	M9 through M18
PMO / Project Controls	2	Full project duration
Operations Readiness Lead + team	2-3	M12 through go-live and 90-day post-go-live support

21A.2 Transition to Operations Staffing Ramp

Operations staffing (Part V, Section 27) ramps ahead of go-live: core Facility Operations Technicians and Security Officers are hired and trained beginning at M14 so the full shift roster is trained and certified on the specific installed systems before Level 5 commissioning (M17), and NOC/DC Operations Engineers are onboarded in parallel to participate in commissioning as an extension of their training program.

22A. Detailed Procurement Schedule and Categories

22A.1 Long-Lead Procurement Categories

Category	Typical Lead Time	Order-By Milestone
Medium-voltage switchgear	9-12 months	M3 (design freeze)
Standby generators	9-14 months	M3 (design freeze)
Chillers	8-12 months	M3 (design freeze)
UPS systems	6-9 months	M3-M4
Structured cabling (bulk fiber/copper)	2-3 months	M9
Network fabric hardware	2-4 months (market dependent)	M10-M11
Compute/storage hardware (Phase 1)	2-4 months (market dependent)	M11-M12

22A.2 Procurement Risk Management

Given the volatility of lead times for switchgear, generators, and certain network/compute hardware categories in constrained supply markets, the PMO maintains a rolling lead-time reforecast every 60 days for all long-lead categories, escalating any category trending beyond the order-by milestone in the table above to the Steering Committee as a schedule risk per Section 23.

23A. Extended Risk Register

Risk	Category	Likelihood	Impact	Mitigation
Geotechnical conditions differ from survey	Site/Civil	Low	Medium	Early geotechnical investigation, design contingency
Key trade contractor insolvency	Construction	Low	High	Pre-qualification, payment bonding
Network/compute hardware chip shortage	IT Procurement	Medium	Medium	Multiple qualified vendors, early order placement
Commissioning reveals design/installation defect	Commissioning	Medium	High	Staged Level 1-4 testing to catch defects early
Operations staff hiring/training delay	Operations Readiness	Low	Medium	Early recruitment start (M12), phased training plan
Utility rate structure changes during delivery	Financial	Low	Medium	Rate lock negotiation where available, TCO sensitivity model
Cybersecurity control gap found in independent audit	Security	Medium	High	Early security design review, pre-go-live pentest

24A. Quality Gate Checklist by Design Stage

Gate	Key Quality Checks
30% Design Review	Concept validated against functional requirements (Section 4A.1); Tier III strategy confirmed feasible
60% Design Review	Discipline coordination / clash detection; long-lead equipment specifications finalized
90% Design Review	Full construction documents issued for permitting; cost estimate reconciled to budget
100% Design / Design Freeze	All disciplines sign off; baseline established for change control (Section 26.2)
Pre-Commissioning Readiness	Punch list from construction closed; test scripts approved by commissioning agent

21B. Detailed Budget Line Items (Illustrative, Phase 1)

21B.1 Electrical Infrastructure Line Items

Line Item	% of Electrical Budget
Medium-voltage switchgear and utility interconnect	22%
UPS systems (modules, batteries/energy storage)	28%
Standby generators and fuel system	24%
Low-voltage distribution (PDUs, RPPs, rPDUs)	14%
Grounding, bonding, surge protection	5%
Installation labor and testing	7%

21B.2 Mechanical Infrastructure Line Items

Line Item	% of Mechanical Budget
Chiller plant	35%
CRAH / in-row cooling units	25%
Piping, pumps, and controls	18%
Containment systems	8%
BMS integration	7%
Installation labor and testing	7%

21B.3 IT and Network Infrastructure Line Items (Phase 1 Payload)

Line Item	% of IT/Network Budget
Compute hardware	30%
Storage hardware	22%
Network hardware (spine/leaf/border/security)	20%
Structured cabling	8%
Racks and power distribution (rack level)	8%
DCIM/monitoring/software licensing	7%
Installation labor and testing	5%

25A. Communication Templates and Escalation Detail

25A.1 Status Report Template Structure

- Executive summary (overall RAG status: red/amber/green).
- Schedule status against milestones (Section 20.2), with SPI where applicable (Section 21.3).
- Budget status against baseline, with CPI (Section 21.3) and variance explanation for categories over 5%.
- Top risks and issues, with owner and target closure date (Section 23).
- Upcoming decisions or approvals needed from the Steering Committee.

25A.2 Escalation Path

Issue Type	First Escalation	Second Escalation
Schedule risk to a milestone	Discipline Lead → Project Director	Project Director → Steering Committee
Budget variance beyond contingency	PMO → Project Director	Project Director → Steering Committee
Safety incident	Site Safety Officer → Construction Manager	Construction Manager → Project Director (immediate)
Design/Tier-rating impacting issue	Facility Design Lead → Project Director	Project Director → Steering Committee + Commissioning Agent

26A. Lessons Learned Process

Lessons learned are captured continuously through a lightweight log maintained by the PMO (not deferred solely to a project-end retrospective), reviewed at each monthly Steering Committee meeting for any process or design lesson that should be applied to a subsequent phase (Section 20.4) or work package still in progress. A formal lessons-learned workshop is held at the close of each major stage gate (Section 24.3) and at Project close, with findings archived alongside the document baseline (Section 26.2) for reference on future projects.

18A. Extended Project Organization Detail

18A.1 RACI at the Steering Committee Level

Decision	Executive Sponsor	Project Director	Discipline Leads
Approve Phase 1 budget baseline	A	R	C
Approve design freeze (Section 20.2)	I	A	R
Approve change > \$250,000 (Section 26.1)	A	R	C
Approve go-live readiness (Appendix K)	A	R	C

18A.2 Meeting Governance

Every governance meeting defined in Section 25.1 operates against a standing agenda template, with minutes and action items logged in the project document management system and carried forward until closed, ensuring decisions and open items are not lost between meetings across the 18-month delivery schedule.

19A. Detailed Work Package Decomposition

19A.1 Site and Facility Shell - Sub-Packages

- 1.1 Geotechnical investigation and site survey
- 1.2 Civil works: grading, drainage, utility trenching
- 1.3 Foundation and structural slab
- 1.4 Structural shell erection
- 1.5 Building envelope: roofing, weatherproofing, cladding
- 1.6 Interior fit-out shell: raised floor, wall partitions, gray-space rooms

19A.2 Electrical Infrastructure - Sub-Packages

- 2.1 Utility interconnect and medium-voltage switchgear
- 2.2 Standby generation and fuel system
- 2.3 UPS systems and battery/energy storage
- 2.4 Low-voltage distribution: PDUs, RPPs, rPDUs
- 2.5 Grounding, bonding, and surge protection
- 2.6 Electrical commissioning (Level 1-3)

19A.3 Mechanical Infrastructure - Sub-Packages

- 3.1 Chiller plant and yard equipment
- 3.2 Piping, pumps, and chilled water distribution
- 3.3 CRAH/in-row cooling units and containment
- 3.4 BMS controls integration
- 3.5 Mechanical commissioning (Level 1-3)

19A.4 Fire and Security Systems - Sub-Packages

- 4.1 VESDA and fire detection
- 4.2 Clean-agent suppression system
- 4.3 Access control and mantrap systems
- 4.4 CCTV and security operations console
- 4.5 Fire and security commissioning

19A.5 IT and Network Infrastructure - Sub-Packages

- 5.1 Racks, cable pathways, and rack-level power distribution
- 5.2 Structured cabling installation and certification

- 5.3 Network fabric hardware and configuration
- 5.4 Compute and storage hardware (Phase 1 payload)
- 5.5 DCIM, monitoring, and observability stack deployment
- 5.6 IT/network commissioning (Level 3-4)

19A.6 Commissioning and Transition - Sub-Packages

- 6.1 Level 4 integrated systems testing
- 6.2 Level 5 full integrated systems test (failure scenario simulation)
- 6.3 SOP/MOP development and validation
- 6.4 Operations staffing, training, and certification
- 6.5 Operational readiness review and go-live

22B. Extended Vendor Management Detail

22B.1 Vendor Onboarding Process

New vendors, once selected per the criteria in Section 22.2, complete a formal onboarding process including contract execution, insurance and safety compliance verification, security/background check requirements for personnel requiring site access (Section 9D.2), and orientation to site-specific safety and MOP requirements (Section 27C.1) before being authorized to perform work.

22B.2 Multi-Vendor Coordination

Where multiple vendors' work is interdependent (e.g., electrical and mechanical contractors coordinating on BMS integration points, Section 9K.1), the Construction Manager maintains an interface control document identifying every cross-vendor dependency, its owner, and its resolution status, reviewed at the weekly site coordination meeting (Section 25.1).

23B. Risk Response Strategy Detail

23B.1 Risk Response Types

Response Type	Description	Example
Avoid	Change plan to eliminate the risk entirely	Select site with confirmed dual utility feed to avoid single-feed risk
Mitigate	Reduce likelihood or impact	Fixed-price contracts to mitigate cost escalation risk
Transfer	Shift risk to a third party (contractually)	Liquidated damages clauses transferring schedule risk to contractor
Accept	Retain the risk with contingency reserve	Management contingency reserve (Section 21.1) for residual risk

24B. Independent Commissioning Agent Role

An independent, third-party commissioning agent (separate from the design and construction teams) oversees Level 4-5 integrated systems testing (Section 20.3, Appendix I) to provide an objective verification that the as-built facility meets the Tier III concurrent-maintainability objective (Section 4.4) and the functional requirements traceability matrix (Section 4A.1), with authority to require remediation of any deficiency found during testing before go-live sign-off (Appendix K) is granted.

26B. Document Control and Version Management

All design, commissioning, and operational documents referenced throughout this document (SOPs/MOPs, as-built drawings, risk register, schedule, budget) are maintained in a single version-controlled project document management system, with access permissions aligned to the RACI structure in Section 18A.1 and a defined naming/versioning convention consistent with the revision history format shown in Appendix L.

20B. Extended Milestone and Task Reference (Representative Detail)

Task	Start	End	Work Package
Kickoff and charter approval	M1 W1	M1 W1	Governance
Site selection finalized	M1 W2	M1 W4	Site & Facility Shell
Geotechnical investigation	M1 W3	M2 W2	Site & Facility Shell
30% design review	M2 W2	M2 W2	All disciplines
60% design review	M2 W4	M2 W4	All disciplines
90% design review	M3 W2	M3 W2	All disciplines
Design freeze / 100% design	M3 W4	M3 W4	Governance
Long-lead equipment orders placed	M3 W4	M4 W1	Electrical / Mechanical
Permitting submission	M3 W4	M4 W4	Site & Facility Shell
Site mobilization	M4 W1	M4 W1	Site & Facility Shell
Foundation complete	M5 W2	M5 W2	Site & Facility Shell
Structural shell complete	M8 W2	M8 W2	Site & Facility Shell
Building weathertight	M9 W1	M9 W1	Site & Facility Shell
Switchgear installed	M9 W4	M10 W2	Electrical Infrastructure
Generator plant installed	M9 W4	M11 W1	Electrical Infrastructure
Chiller plant installed	M9 W4	M11 W1	Mechanical Infrastructure
UPS installed	M10 W3	M11 W4	Electrical Infrastructure
CRAH units installed	M10 W3	M11 W4	Mechanical Infrastructure
Containment installed	M11 W1	M11 W4	Mechanical Infrastructure
Electrical energization	M12 W1	M12 W4	Electrical Infrastructure
Mechanical plant operational	M12 W1	M12 W4	Mechanical Infrastructure
Racks and pathways installed	M11 W1	M11 W4	IT & Network Infrastructure
Structured cabling complete/certified	M12 W1	M13 W1	IT & Network Infrastructure
Network fabric installed and configured	M13 W1	M13 W4	IT & Network Infrastructure
Compute/storage installed (Phase 1)	M14 W1	M14 W4	IT & Network Infrastructure
DCIM/monitoring deployed	M14 W1	M14 W3	IT & Network

Task	Start	End	Work Package
			Infrastructure
Level 1-2 commissioning complete	M13 W1	M15 W2	Commissioning
Level 3 commissioning complete	M15 W1	M15 W4	Commissioning
Level 4 commissioning complete	M16 W1	M16 W3	Commissioning
Level 5 commissioning complete	M16 W4	M17 W2	Commissioning
Operations staff trained/certified	M14 W1	M17 W2	Commissioning & Transition
Operational readiness review	M17 W3	M17 W4	Commissioning & Transition
Go-live	M18 W1	M18 W1	Governance

21C. Extended Cost Category Reference (Full Breakdown, Illustrative)

Category	Sub-Category	% of Total Phase 1 CapEx
Shell, civil, structural	Site prep and civil works	8%
Shell, civil, structural	Structural shell and envelope	10%
Shell, civil, structural	Interior fit-out shell	4%
Electrical	MV switchgear and utility interconnect	6.6%
Electrical	UPS and battery/energy storage	8.4%
Electrical	Generators and fuel system	7.2%
Electrical	Low-voltage distribution	4.2%
Electrical	Grounding, bonding, surge protection	1.5%
Electrical	Installation and testing	2.1%
Mechanical	Chiller plant	7%
Mechanical	CRAH/in-row units	5%
Mechanical	Piping, pumps, controls	3.6%
Mechanical	Containment	1.6%
Mechanical	BMS integration	1.4%
Mechanical	Installation and testing	1.4%
Fire and Security	Detection and suppression	2.5%
Fire and Security	Access control and CCTV	1.9%
Fire and Security	Installation and testing	0.6%
IT/Network	Compute hardware	4.8%
IT/Network	Storage hardware	3.5%
IT/Network	Network hardware	3.2%
IT/Network	Cabling, racks, power distribution	2.6%
IT/Network	DCIM/monitoring/software	1.1%
IT/Network	Installation and testing	0.8%
Fees	Design, commissioning, PM fees	5%
Contingency	Management + design contingency (Section 21.1)	Held separately, not in above %

25B. Meeting Agenda Templates

25B.1 Steering Committee Standing Agenda

- Review of prior action items
- Overall RAG status and executive summary
- Schedule status against milestones (Section 20.2)
- Budget status and variance review (Section 21.2)
- Top risks requiring Steering Committee attention (Section 23)
- Change requests requiring Steering Committee approval (Section 26.1)
- Decisions needed / next steps

25B.2 Weekly Discipline Lead Coordination Agenda

- Cross-discipline interface issues (Section 22B.2)
- Schedule look-ahead (next 2-4 weeks)
- Open RFIs / design clarifications
- Safety observations
- Action item review

27. Operations Organization and Staffing

27.1 Operating Model

The facility operates on a 24x7x365 staffing model with on-site Facility and Security shifts at all times, and an on-call escalation path to IT/Network engineering for issues beyond the on-site team's authority to resolve. Staffing is sized to support the Phase 1 footprint and scales with each subsequent build-out phase per the staffing plan reviewed at each phase trigger (Section 20.4).

27.2 Roles and Shift Structure

Role	Coverage	Primary Responsibility
Facility Operations Technician	24x7 (shift rotation)	Electrical/mechanical plant monitoring, rounds, MOPs
Security Officer	24x7 (shift rotation)	Access control, CCTV monitoring, incident response
NOC/DC Operations Engineer	24x7 (shift rotation)	IT/network monitoring, incident triage, ticket management
Facility Manager	Business hours + on-call	Overall facility management, vendor coordination
Network/Systems Engineer (on-call)	On-call rotation	Escalation for complex IT/network incidents
DC Operations Manager	Business hours + on-call	Team leadership, SLA and vendor management

27.3 Training and Certification

Facility Operations Technicians complete a structured onboarding program covering the electrical and mechanical systems described in Part II before being authorized to perform unsupervised switching or maintenance operations, with periodic recertification aligned to the maintenance program in Section 30. NOC engineers complete onboarding covering the network, compute, and storage architecture in Part III, plus incident and change management process training.

28. Standard Operating Procedures

28.1 SOP Framework

Standard Operating Procedures (SOPs) and Methods of Procedure (MOPs) govern all routine and non-routine work in the facility. Every SOP/MOP specifies the trigger condition, prerequisites, step-by-step procedure, rollback plan, and required approvals, and is version-controlled in the same document management system as the design baseline (Section 26.2).

28.2 Core SOP Categories

- Routine rounds and inspection SOPs (electrical, mechanical, fire, security systems).
- Planned maintenance MOPs for every major system (Section 30), including risk classification and required sign-offs.
- Emergency response SOPs (utility failure, generator failure, cooling failure, fire alarm, security incident).
- IT operations SOPs (server/network provisioning, decommissioning, patching, backup verification).
- Access control SOPs (visitor escort, vendor access, emergency access override).

28.3 MOP Risk Classification

Risk Class	Description	Approval Required
Low	No potential to impact live IT load if executed correctly	Shift lead sign-off
Medium	Isolated component maintenance with redundancy in place	Facility Manager sign-off, peer review
High	Any activity with potential to impact live load if a step fails	Facility Manager + DC Operations Manager, dry-run/tabletop review

29. Monitoring and Alerting

29.1 Monitoring Architecture

The Network Operations Center (NOC) monitors the facility through an integrated stack combining the DCIM platform (Section 16) for power/cooling/environmental telemetry, a network and systems monitoring platform for compute/storage/network health, and the SIEM (Section 17.4.1) for security events, unified into a single alerting and dashboard layer to give shift engineers a consolidated operational view.

29.2 Alert Severity and Response Targets

Severity	Definition	Response Target	Escalation
P1 - Critical	Live production impact or imminent risk to redundancy	Immediate, 24x7	Auto-page on-call + DC Operations Manager
P2 - High	Redundancy loss without current production impact	< 30 minutes	Auto-page on-call
P3 - Medium	Degraded performance, no redundancy loss	< 4 hours	Ticket assigned, business-hours follow-up
P4 - Low	Informational, no service impact	Next business day	Ticket logged

29.3 Dashboards and Reporting

Real-time dashboards display current PUE, critical load, redundancy status (N/N+1/2N state of each system), and open incident counts, visible in the NOC and available to facility leadership. A daily operations summary and a monthly executive summary are generated automatically from the monitoring stack.

30. Preventive Maintenance

30.1 Maintenance Philosophy

Preventive maintenance follows manufacturer-recommended intervals as a floor, adjusted upward in frequency for systems identified as higher-risk through reliability-centered maintenance (RCM) analysis. All maintenance on redundant systems is scheduled and executed one component/path at a time, never taking both the A and B path (or N and N+1 unit) out of service simultaneously.

30.2 Maintenance Schedule (Representative)

System	Activity	Frequency
UPS systems	Full inspection and battery testing	Quarterly
Standby generators	Load-bank test at full rated load	Annually (rotating unit, N+1 coverage maintained)
Standby generators	No-load exercise run	Weekly
Switchgear	Infrared thermography inspection	Quarterly
Chillers / CRAH units	Full mechanical service	Semi-annually
Fire detection (VESDA)	Sensitivity testing and calibration	Monthly
Fire suppression system	Full discharge test (rotating zone)	Annually
Access control / CCTV	Functional test of all devices	Quarterly
Structured cabling	Fiber loss/OTDR spot audit	Annually

30.3 Maintenance Records

All maintenance activity is logged against the corresponding DCIM asset record (Section 16.2), providing a complete maintenance history per device used both for warranty/vendor performance tracking and for reliability-centered maintenance interval adjustments.

31. Incident and Problem Management

31.1 Incident Management Process

Incidents are logged, triaged, and tracked through the ITSM platform using the severity model in Section 29.2. Every P1 incident triggers an incident commander role (typically the on-call DC Operations Manager) responsible for coordinating response, communication, and post-incident review, independent of the engineer performing hands-on remediation.

31.2 Post-Incident Review

Every P1 and P2 incident is followed by a formal post-incident review within five business days, producing a root cause analysis and a set of corrective actions with owners and target dates, tracked to closure through the problem management backlog. Recurring root causes trigger a mandatory reliability engineering review.

31.3 Problem Management

Problem management maintains a known-error database linking incidents to underlying root causes and workarounds, reviewed monthly to identify systemic issues (e.g., a specific hardware batch, a recurring firmware bug) that warrant proactive remediation ahead of further incidents.

32. Business Continuity and Disaster Recovery

32.1 BC/DR Strategy

Business continuity planning addresses continued facility operation through partial system failures (covered by the N+1/2N redundancy design in Part II) and disaster recovery addresses loss of the entire facility, relying on the replication architecture described in Section 12.4 to a secondary site or region.

32.2 Recovery Objectives

Workload Tier	RTO	RPO
Tier 1 - Critical production	< 4 hours	< 15 minutes (near-sync replication)
Tier 2 - Standard production	< 24 hours	< 4 hours
Tier 3 - Non-production	Best effort	Best effort

32.3 DR Testing

Disaster recovery capability is tested through scheduled failover exercises at least annually for Tier 1 workloads, including a full or partial cutover to the secondary site, with results and any gaps documented and remediated ahead of the next test cycle. Facility-level failure scenarios (utility loss, generator failure, cooling failure) are exercised as part of the Level 5 commissioning program (Section 20.3) and subsequently through periodic tabletop and live-failover drills.

33. Capacity Planning

33.1 Capacity Management Process

Capacity across power, cooling, space, network ports, compute, and storage is tracked continuously in the DCIM and monitoring platforms, with a monthly capacity review comparing current utilization and growth trend against available headroom in the current phase, feeding the phase-trigger decision described in Section 20.4.

33.2 Capacity Thresholds and Actions

Utilization	Action
< 60%	Monitor, no action required
60-75%	Begin planning for next capacity increment
75-90%	Initiate procurement for next phase / capacity increment (trigger point, Section 20.4)
> 90%	Escalate to Steering Committee, evaluate interim measures

34. Change and Configuration Management

34.1 Operational Change Management

All post-go-live changes to the facility's electrical, mechanical, network, or IT configuration follow a change management process modeled on the project-phase process in Section 26, scaled for operational cadence: standard (pre-approved, low-risk), normal (reviewed by a Change Advisory Board), and emergency (expedited approval for incident remediation, retrospectively reviewed).

34.2 Configuration Management Database

The DCIM platform and IT asset/configuration management database (CMDB) together form the authoritative configuration record, updated as a mandatory step in the change closure process; a change is not considered complete until the corresponding DCIM/CMDB records are updated.

35. Vendor and SLA Management

35.1 Vendor Governance

Ongoing maintenance and support vendors (UPS/generator service, mechanical service, network/hardware support, security systems) are managed against defined SLAs for response time and resolution time, tiered by the severity model in Section 29.2, with quarterly vendor performance reviews.

35.2 SLA Targets by Vendor Category

Vendor Category	P1 Response	P1 Onsite (if required)
Electrical (UPS/generator/switchgear)	< 30 minutes (remote)	< 4 hours
Mechanical (chiller/CRAH)	< 30 minutes (remote)	< 4 hours
Network/compute hardware	< 30 minutes (remote)	< 4 hours (next business day for non-P1)
Fire and security systems	< 1 hour (remote)	< 8 hours

36. Sustainability and Energy Management

36.1 Energy Management Program

The facility operates an energy management program aligned to ISO 50001, tracking PUE and WUE continuously (Section 7.4) and reviewing efficiency opportunities (e.g., expanded economizer/free-cooling hours, containment optimization, set-point adjustments within the ASHRAE allowable envelope) on a quarterly basis.

36.2 Renewable Energy and Carbon Reporting

Where available in the local utility market, the facility procures renewable energy credits or enters power purchase agreements to reduce the carbon intensity of its utility power supply, with progress tracked and reported alongside PUE/WUE in the quarterly sustainability report referenced in Section 25.2.

27A. Shift Operations Detail

27A.1 Shift Handover Process

Every shift change includes a structured handover covering open incidents, in-progress maintenance activities (with MOP reference), any systems currently operating in a degraded-redundancy state, and any scheduled work expected during the incoming shift. Handover is documented in the shift log within the ITSM platform and verbally reviewed between outgoing and incoming shift leads before the outgoing shift departs.

27A.2 Rounds and Inspection Schedule

Round Type	Frequency	Scope
Electrical room walk-through	Every shift	Visual inspection, abnormal noise/smell/heat check, panel status
Mechanical plant walk-through	Every shift	Chiller/CRAH status, leak detection panel check
Data hall environmental spot-check	Every shift	Hot/cold aisle temperature spot-check, containment integrity
Generator yard inspection	Daily	Fuel level, block heater status, visual inspection
Fire/life safety panel check	Daily	Panel status, no active trouble alarms
Full facility security sweep	Every shift	Perimeter, entrances, CCTV coverage verification

27A.3 Staffing Growth by Phase

Phase	Critical IT Load	Facility Ops Technicians (per shift)	NOC Engineers (per shift)	Security Officers (per shift)
Phase 1	1.0 MW	2	2	2
Phase 2	2.0 MW	3	3	2
Phase 3	3.0 MW	3	3	3
Phase 4	4.0 MW	4	4	3

28A. Representative Detailed SOPs

28A.1 SOP Example: Planned UPS Module Maintenance (Medium Risk)

Trigger: Scheduled quarterly UPS maintenance (Section 30.2) or vendor-identified module fault requiring replacement. Prerequisites: confirm B-path UPS is healthy and carrying no faults; confirm no other maintenance activity is in progress on the B path; notify NOC and log a maintenance change record. Procedure: place the target A-path module into maintenance bypass per vendor MOP, verify downstream load remains stable on the redundant module(s) within the same path, perform the physical maintenance/replacement, run vendor-specified post-maintenance diagnostics, return the module to service, and confirm the A path returns to full N+1 module redundancy before closing the change. Rollback: if downstream load shows any instability after placing the module into bypass, immediately abort and return the module to normal operation before proceeding further.

28A.2 SOP Example: Data Hall Temperature Excursion Response

Trigger: DCIM/BMS alarm indicating a rack or row temperature reading outside the ASHRAE allowable envelope (Section 7.2.1). Procedure: shift engineer confirms the alarm is not a sensor fault via a secondary reading, checks containment integrity (doors closed, blanking panels present) in the affected row, checks CRAH/in-row unit status for the affected zone, and escalates to the on-call mechanical vendor per the SLA in Section 35.2 if the excursion is not resolved by an operator-correctable cause within 15 minutes. All temperature excursion events are logged and reviewed in the monthly capacity and reliability review.

28A.3 SOP Example: Emergency Access Override

Trigger: Life-safety emergency requiring immediate facility access that cannot wait for standard credential verification. Procedure: emergency access override is authorized only by the on-site shift lead or DC Operations Manager, is logged automatically by the access control system with the authorizing individual's credentials, and triggers a mandatory post-event review within 24 hours to confirm the override was appropriately used.

29A. Extended Monitoring Detail

29A.1 Key Monitored Metrics by Domain

Domain	Representative Metrics Monitored
Electrical	Utility voltage/current, UPS load %, UPS battery health, generator status, PDU circuit load
Mechanical	Supply/return air temperature, chilled water supply/return temp, chiller load %, humidity
Network	Interface utilization, error/discard rates, BGP session state, latency between racks/sites
Compute/Storage	CPU/memory utilization, storage latency/IOPS, capacity utilization, hardware health (SMART, ECC errors)
Security	Failed authentication attempts, firewall policy violations, badge/mantrap anomalies
Environmental	Rack-level temperature/humidity, water leak detection, smoke detection status

29A.2 Alert Tuning and Noise Reduction

Alert thresholds are reviewed quarterly to balance early detection against alert fatigue; alerts that are frequently acknowledged without corrective action are investigated for threshold tuning, correlation/suppression rules, or underlying root-cause remediation, tracked through the same problem management process referenced in Section 31.3.

32A. Extended Disaster Recovery Detail

32A.1 DR Site Architecture

Where a secondary DR site or region is in scope, its infrastructure mirrors the segmentation and security model of the primary facility (Section 13.2, Section 17) at a scale sized to the Tier 1/Tier 2 workload footprint requiring failover capability, rather than a full mirror of total facility capacity. Network connectivity between primary and DR sites uses redundant, diversely routed circuits consistent with the WAN redundancy principles in Section 13.6.1.

32A.2 Failover Runbook Structure

Each Tier 1 application maintains a documented failover runbook specifying pre-failover validation checks, the failover execution sequence (including DNS/traffic redirection steps), post-failover validation checks, and the corresponding failback procedure, version-controlled alongside the SOP/MOP library (Section 28.1).

32A.3 Communication During a DR Event

A DR event triggers the same incident commander model used for P1 incidents (Section 31.1), with a dedicated communication track to business stakeholders separate from the technical execution track, ensuring technical responders are not interrupted for status updates during active failover execution.

33A. Extended Capacity Planning Detail

33A.1 Capacity Planning Inputs

- Historical utilization trend (power, cooling, compute, storage, network port) from the DCIM and monitoring platforms.
- Forward-looking demand forecast from application/business owners, reconciled quarterly against actuals.
- Hardware refresh cycles (Section 11.2) which affect both capacity and the mix of workload that can be hosted per rack.
- Lead times for the relevant capacity category (Section 22A.1) which set how far in advance a capacity trigger must be acted upon.

33A.2 Capacity Reporting

A monthly capacity report is distributed to the PMO and Steering Committee summarizing current utilization against each capacity dimension (power, cooling, space, network, compute, storage), trend versus the prior quarter, and projected date to reach the next action threshold defined in Section 33.2.

36A. Extended Sustainability Program Detail

36A.1 Waste Heat Reuse

Where site conditions and nearby demand allow, the mechanical design evaluates waste heat reuse (e.g., district heating, adjacent facility heating) as a further efficiency measure beyond economizer cooling (Section 9B.2), assessed on a site-specific cost-benefit basis during detailed mechanical design.

36A.2 Water Stewardship

Beyond the WUE target in Section 4.4, the facility evaluates cooling tower water treatment and cycles-of-concentration optimization to reduce blowdown water consumption, and where feasible, uses reclaimed or non-potable water sources for cooling tower makeup to reduce potable water demand.

36A.3 Sustainability Reporting Cadence

Report	Frequency	Audience
PUE/WUE operational dashboard	Real-time / continuous	NOC, Facility Manager
Efficiency opportunity review	Quarterly	DC Operations Manager, Facility Manager
Sustainability summary (PUE, WUE, renewable %)	Quarterly	Steering Committee
Annual sustainability report	Annually	Executive sponsors, external reporting as applicable

27B. Training Curriculum Detail

27B.1 Facility Operations Technician Curriculum

Module	Topics	Duration
Electrical systems fundamentals	Utility/switchgear/UPS/generator architecture (Part II, Section 6)	1 week
Mechanical systems fundamentals	Chiller plant, CRAH/in-row, containment (Part II, Section 7)	1 week
Fire and life safety systems	Detection, suppression, EPO (Part II, Section 8, 9A.5)	3 days
MOP execution and risk classification	SOP/MOP framework (Section 28)	3 days
DCIM and monitoring platform	Alarm response, dashboard interpretation (Section 16, 29)	2 days
Site-specific shadow shifts	Supervised on-shift practice before independent certification	4 weeks

27B.2 NOC/DC Operations Engineer Curriculum

Module	Topics	Duration
Network and compute/storage architecture	Part III overview and detailed design	2 weeks
Monitoring and observability platform	Section 29, 17I	1 week
Incident and problem management process	Section 31	3 days
Change management process	Section 34	2 days
Security operations basics	Section 17D.3, 17.4.1	3 days
Site-specific shadow shifts	Supervised on-shift practice before independent certification	4 weeks

27B.3 Recertification

All operations staff complete an annual recertification covering updates to SOPs/MOPs, any facility modifications since the prior certification, and a practical assessment for high-risk MOP categories (Section 28.3), consistent with the maintenance and training cadence used for life-safety systems.

27C. Environmental Health and Safety Program

27C.1 Workplace Safety

The facility operates a documented EHS program covering arc-flash safety (per NFPA 70E) for all electrical work, lockout-tagout procedures for mechanical maintenance, confined-space procedures where applicable, and personal protective equipment (PPE) requirements by task, integrated into the MOP framework (Section 28) so PPE and safety procedure requirements are specified as part of every relevant MOP.

27C.2 Incident and Near-Miss Reporting

All safety incidents and near-misses are reported and investigated through a dedicated safety incident process, tracked separately from operational incidents (Section 31) but reviewed at the same monthly operations review cadence, with corrective actions tracked to closure.

27C.3 Emergency Response Planning

The facility maintains an emergency response plan covering severe weather, seismic events (where applicable to the site), medical emergencies, and security threats, with defined roles, muster points, and communication procedures, exercised through periodic drills coordinated with local emergency services where appropriate.

31A. Extended Incident Management Detail

31A.1 Incident Classification Examples

Scenario	Severity	Rationale
Single UPS module fault, N+1 maintained	P2	Redundancy loss without production impact
Both A and B network leaf switches down for one rack	P1	Live production impact
Single generator fault during monthly test, N+1 maintained	P3	No redundancy loss during normal utility operation
Utility outage with successful generator transfer	P2	No production impact, but reduced redundancy margin
Confirmed unauthorized data hall access attempt	P1	Security-critical, immediate response required

31A.2 Communication During Major Incidents

Major incidents (P1) trigger a defined communication cadence to stakeholders: initial notification within 15 minutes of confirmation, status updates at a minimum of every 30 minutes until resolution, and a final resolution notice with a preliminary root-cause summary within 24 hours, ahead of the full post-incident review described in Section 31.2.

35A. Vendor Contract Management Detail

35A.1 Contract Renewal and Performance Review Cycle

Maintenance and support contracts are reviewed no later than 90 days before renewal, incorporating the quarterly vendor performance scorecards (response time adherence, resolution time adherence, quality of work) into the renewal negotiation, with underperforming vendors placed on a formal improvement plan before renewal is considered.

35A.2 Spare Parts and Critical Spares Strategy

Critical spares for long-lead or safety-critical components (UPS power modules, generator control boards, key network line cards) are held on-site or under a guaranteed-delivery vendor agreement, sized against the MTTR targets in Section 1.1 and the SLA targets in Section 35.2, rather than relying solely on vendor stock availability during a P1 incident.

Operations Audit and Reporting Calendar

Activity	Frequency	Owner
Access recertification review	Quarterly	Security Lead
Firewall rule-base review	Quarterly	Security Lead
Capacity report	Monthly	DC Operations Manager
Vendor performance scorecard	Quarterly	DC Operations Manager
DR failover test (Tier 1 workloads)	Annually	Operations Readiness Lead
Full facility fire/life safety inspection	Annually (3rd-party)	Facility Manager
Independent penetration test	Annually	Security Lead
Sustainability report	Quarterly / Annually	Facility Manager
Operational readiness / Tier compliance self-audit	Annually	DC Operations Manager

27D. Continuous Improvement Program

27D.1 Operational Metrics Review

Beyond incident-driven post-incident reviews (Section 31.2), the operations team conducts a monthly continuous improvement review examining trend data across MTTR, alert volume and false-positive rate, MOP execution issues, and near-miss safety reports (Section 27C.2), identifying process or design improvements independent of any single incident.

27D.2 SOP/MOP Currency Review

Every SOP and MOP is reviewed for currency at least annually, or immediately following any facility modification that affects the procedure, with the review confirming the procedure still matches the as-built configuration recorded in the DCIM/CMDB (Section 16) and reflects current best practice for the system it covers.

27E. On-Call and Escalation Management

27E.1 On-Call Rotation Structure

On-call rotations for NOC/DC Operations Engineers and Facility/Mechanical/Electrical escalation contacts follow a documented weekly rotation with a minimum of two qualified engineers available at any time (primary and secondary), published in advance and integrated with the paging system referenced in the alert severity model (Section 29.2).

27E.2 On-Call Load Management

On-call page volume and after-hours response time are tracked as an operational health metric; sustained high on-call burden for a specific system is treated as a signal for the continuous improvement review (Section 27D.1) to investigate underlying reliability or alert-tuning issues rather than being accepted as a normal cost of operations.

28B. Change Freeze Periods

Standard (non-emergency) changes to production systems are restricted during defined change-freeze periods around high-risk calendar events (e.g., known peak business demand periods) to reduce the likelihood of a self-inflicted incident during a period when the business impact of an outage would be highest; freeze periods are published at least one quarter in advance and exceptions require Facility Manager or DC Operations Manager approval per Section 34.1.

29B. Reporting and Analytics for Leadership

29B.1 Executive Operations Dashboard

A summarized executive dashboard, distinct from the detailed NOC operational dashboards (Section 29.3), presents trailing 12-month availability, PUE/WUE trend, open critical risk items, and capacity runway against the phase triggers in Section 20.4, refreshed monthly for the Steering Committee and, post-handover, for ongoing executive governance of the operating facility.

29B.2 Availability Reporting Methodology

Availability is calculated and reported using a consistent methodology defined at project start: measured at the IT load (rack PDU) level rather than only at the utility meter, with any redundancy-loss event logged and tracked as a distinct metric from an actual production-impacting outage, so the organization can distinguish true availability performance from the frequency of degraded-redundancy events that did not reach the IT load.

32B. Extended Business Continuity Scope

32B.1 Non-IT Business Continuity Considerations

Beyond IT/data disaster recovery (Section 32), the facility's business continuity plan addresses continuity of the operations team itself: cross-training so no single individual is a single point of knowledge failure (Section 27B), a documented succession approach for key roles (Facility Manager, DC Operations Manager), and a remote/alternate work arrangement for functions that do not strictly require physical presence during a facility-access disruption that does not itself constitute an IT outage.

32B.2 Supplier Continuity

Critical vendor relationships (Section 35) are assessed for the vendor's own continuity risk (e.g., a sole-source spare-parts supplier), with alternate qualified vendors identified for the highest-criticality categories consistent with the supply-chain risk approach described in Section 17D.5.

33B. Long-Term Capacity and Refresh Planning

33B.1 Hardware Refresh Cadence

Compute, storage, and network hardware refresh follows the lifecycle referenced in Section 11.2 (4-5 years for compute), planned and budgeted as a recurring OpEx/minor-CapEx program rather than an unplanned event, with refresh waves staggered across racks/rows to avoid a single large disruptive cutover and to naturally introduce new hardware generations' efficiency gains into the PUE trend over time.

33B.2 End-of-Life Risk Management

Hardware and software nearing vendor end-of-support is tracked in the CMDB with automated alerting well ahead of the support expiry date, feeding the refresh planning cycle so no production system runs on unsupported hardware or software without a documented, risk-accepted exception approved at the DC Operations Manager level.

28C. Complete SOP/MOP Library Index (Representative)

SOP/MOP ID	Title	Risk Class
EL-001	Routine electrical room inspection round	Low
EL-010	Planned UPS module maintenance	Medium
EL-011	UPS full system load-bank test	High
EL-020	Generator monthly no-load exercise	Low
EL-021	Generator annual full load-bank test	High
EL-030	Switchgear infrared thermography inspection	Low
EL-031	Planned switchgear breaker maintenance	High
ME-001	Routine mechanical plant inspection round	Low
ME-010	Planned chiller maintenance (single unit)	Medium
ME-020	CRAH/in-row unit filter and coil service	Low
ME-030	Containment integrity check and repair	Low
FS-001	VESDA sensitivity test and calibration	Medium
FS-010	Clean-agent system annual discharge test	High
SEC-001	Visitor/vendor escort procedure	Low
SEC-010	Emergency access override	High
SEC-020	Access control system planned maintenance	Medium
NET-001	Planned network device firmware update	Medium
NET-010	Single leaf switch replacement	Medium
NET-020	Spine switch planned maintenance	High
IT-001	Server provisioning (standard build)	Low
IT-010	Server decommissioning and secure erasure	Medium
IT-020	Storage array planned firmware update	High
DR-001	Tier 1 application DR failover test	High

29C. Detailed Escalation Timing Reference

Time Since Detection	Expected State (P1 Incident)
0-5 minutes	Auto-page fired, on-call engineer acknowledges
5-15 minutes	Initial triage complete, incident commander assigned, stakeholder notification sent
15-30 minutes	Remediation action identified and in progress, or vendor engaged per SLA (Section 35.2)
30-60 minutes	Status update sent; if unresolved, DC Operations Manager engaged directly
Every 30 minutes thereafter	Status update sent per Section 31A.2 until resolution
Within 24 hours of resolution	Preliminary root-cause summary issued
Within 5 business days	Formal post-incident review completed (Section 31.2)

30A. Extended Maintenance Reference Schedule

System	Activity	Frequency
Batteries / energy storage	Impedance/capacity testing	Annually
Automatic transfer switches	Functional test under load	Annually
Grounding system	Ground resistance testing	Annually
Surge protective devices	Functional/status check	Quarterly
BMS controllers	Backup/failover test	Annually
Leak detection cable	Functional test at multiple points	Semi-annually
Emergency lighting / egress signage	Functional test	Monthly
DCIM/monitoring platform	Backup and disaster recovery test of the platform itself	Annually
IPAM/DNS/DHCP infrastructure	Redundancy failover test	Annually
Certificate authority / PKI	Root/intermediate CA key ceremony audit	Annually

31B. Problem Management Categories

Category	Example	Typical Owner
Hardware defect (batch/firmware)	Recurring NIC firmware crash across a hardware batch	IT/Network Architecture Lead
Design gap	Cooling hot spot not resolved by tuning alone	Facility Design Lead
Process gap	MOP missing a required verification step	DC Operations Manager
Capacity-driven	Recurring alerts caused by approaching capacity threshold	DC Operations Manager (feeds Section 33)
Vendor performance	Repeated SLA miss by a maintenance vendor	DC Operations Manager (feeds Section 35A.1)

35B. Extended SLA Reference by System

System	P2 Response	P3 Response	Notes
Electrical (UPS/generator/switchgear)	< 2 hours	Next business day	Section 9A/6
Mechanical (chiller/CRAH)	< 2 hours	Next business day	Section 7/9B
Network/compute hardware	< 2 hours	Next business day	Section 11-13
Fire and security systems	< 4 hours	2 business days	Section 8-9
DCIM/monitoring software	< 4 hours (remote)	3 business days	Section 16, 29

Appendix A: Glossary of Terms

Term	Definition
2N	Fully redundant topology; two complete, independent systems each capable of carrying full load
BMS	Building Management System - controls and monitors mechanical/electrical building systems
CRAH	Computer Room Air Handler - chilled-water based air handling unit
DCIM	Data Center Infrastructure Management - software platform for physical infrastructure management
EVPN	Ethernet VPN - control plane protocol for VXLAN overlay networks
N+1	Redundancy level with one additional unit beyond the minimum required (N) to carry load
NOC	Network Operations Center
PDU	Power Distribution Unit
PUE	Power Usage Effectiveness - ratio of total facility power to IT equipment power
RPO	Recovery Point Objective - maximum tolerable data loss, measured in time
RTO	Recovery Time Objective - maximum tolerable time to restore service
SIEM	Security Information and Event Management
SLA	Service Level Agreement
Tier (Uptime Institute)	Classification (I-IV) of data center infrastructure redundancy and maintainability
VESDA	Very Early Smoke Detection Apparatus - aspirating smoke detection system
VXLAN	Virtual Extensible LAN - overlay network encapsulation protocol
WUE	Water Usage Effectiveness - ratio of water used to IT equipment energy consumed

Appendix B: Reference Standards

Standard	Title / Scope
Uptime Institute Tier Standard	Topology and Operational Sustainability classification
TIA-942	Telecommunications Infrastructure Standard for Data Centers
BICSI 002	Data Center Design and Implementation Best Practices
ASHRAE TC 9.9	Thermal Guidelines for Data Processing Environments
NFPA 75	Standard for Fire Protection of Information Technology Equipment
NFPA 76	Standard for Fire Protection of Telecommunications Facilities
NFPA 2001	Standard on Clean Agent Fire Extinguishing Systems
ISO/IEC 27001	Information Security Management Systems
ISO 50001	Energy Management Systems
IEEE 1100	Recommended Practice for Powering and Grounding Electronic Equipment

Appendix C: Sample Commissioning Checklist (Level 4 - Integrated Systems)

- Utility power loss simulated; confirm automatic transfer to generator within design threshold.
- Generator start and stable output confirmed within design threshold; UPS ride-through validated.
- UPS-to-generator transfer confirmed with no interruption to downstream IT load (test load bank).
- Single chiller/CRAH unit removed from service; confirm remaining N+1 capacity maintains design thermal envelope.
- Single UPS module/path removed from service; confirm A/B redundancy maintains full IT load.
- Fire detection cross-zone alarm simulated; confirm suppression release logic and manual abort function.
- Access control and mantrap tested for fail-secure and fail-safe behavior under power loss scenario.
- DCIM alarms confirmed for every simulated fault condition above.
- All punch list items from Level 3 system testing closed prior to Level 4 sign-off.

Appendix D: Sample Method of Procedure (MOP) Template Fields

Field	Description
MOP ID / Version	Unique identifier and version-controlled revision
Risk Classification	Low / Medium / High per Section 28.3
Systems Affected	Specific equipment/circuits impacted
Prerequisites	Conditions that must be true before starting (e.g., redundant path confirmed healthy)
Step-by-Step Procedure	Numbered steps with expected result at each step
Rollback Plan	Steps to safely reverse the procedure if an unexpected condition occurs
Approvals Required	Named sign-off roles per risk classification
Post-Work Verification	Steps to confirm system returned to normal/redundant state

Appendix E: Sample RACI Matrix (Selected Activities)

Activity	Facility Design Lead	IT/Network Lead	Commissioning Mgr	Ops Readiness Lead
Electrical design approval	A/R	C	C	I
Network fabric design approval	I	A/R	C	I
Commissioning test scripts	C	C	A/R	C
SOP/MOP development	C	C	C	A/R
Go-live readiness sign-off	C	C	C	A/R

Legend: A = Accountable, R = Responsible, C = Consulted, I = Informed.

Appendix F: Document Control

Field	Value
Document Title	Data Center Project - Master Technical, Business, and Operations Reference
Classification	Internal - Engineering and Technical Team Distribution
Status	Draft for engineering review
Owner	Project Director / PMO
Review Cycle	Updated at each major design gate and quarterly during operations

Appendix G: Extended Glossary

Term	Definition
ACL	Access Control List - rule set controlling network traffic permit/deny decisions
BGP	Border Gateway Protocol - routing protocol used both for internet routing and as a fabric underlay protocol
Blast Radius	The scope of systems/users impacted by a given failure or change
CMDB	Configuration Management Database - authoritative record of IT assets and their relationships
Clos Topology	A multi-stage switching topology (e.g., spine-leaf) providing non-blocking connectivity
Cold Aisle Containment	Physical enclosure of the cold aisle to prevent mixing of supply and return air
CPTED	Crime Prevention Through Environmental Design - physical security design methodology
DAC	Direct Attach Copper - copper cabling assembly used for short-range, high-speed links
Day Tank	Local fuel tank at a generator, replenished from bulk storage
EVPN	Ethernet VPN - BGP-based control plane for VXLAN overlays
Free Cooling / Economization	Use of favorable ambient conditions to reduce mechanical cooling load
IPAM	IP Address Management - system for tracking and allocating IP address space
ITSM	IT Service Management - discipline and tooling for incident, change, and problem management
KMS	Key Management System - centralized management of cryptographic keys
MOP	Method of Procedure - detailed, approved procedure for executing a specific task
Mantrap	Interlocked double-door vestibule enforcing single-person, verified passage
OTDR	Optical Time-Domain Reflectometer - instrument for testing fiber optic cable integrity
PAM	Privileged Access Management - control of elevated/administrative access credentials
RCM	Reliability-Centered Maintenance - methodology for setting maintenance strategy based on failure risk
RTO/RPO	Recovery Time/Point Objective - time and data-loss recovery targets for DR

Term	Definition
SDN	Software-Defined Networking - centralized, programmatic network control
SPD	Surge Protective Device
SPI/CPI	Schedule/Cost Performance Index - earned value management metrics
STS	Static Transfer Switch - sub-cycle switch between two power sources
VXLAN	Virtual Extensible LAN - Layer 2 overlay tunneling protocol over a Layer 3 network
WAF	Web Application Firewall
Zero Trust	Security model requiring continuous verification rather than implicit trust based on network location

Appendix H: Extended Standards and Reference Codes

Standard / Code	Scope
NFPA 70 (NEC)	National Electrical Code - electrical installation safety
NFPA 70E	Electrical Safety in the Workplace
NFPA 780	Standard for the Installation of Lightning Protection Systems
NFPA 13	Standard for the Installation of Sprinkler Systems
IEEE 493	Recommended Practice for Design of Reliable Industrial and Commercial Power Systems
ANSI/TIA-568	Commercial Building Telecommunications Cabling Standard
ISO 22301	Business Continuity Management Systems
SOC 2 (AICPA)	Trust services criteria commonly required for third-party facility audits
Local Building/Seismic Code	Jurisdiction-specific structural and life-safety requirements
Local Environmental Permitting	Stormwater, emissions (generator), and fuel storage permitting requirements

Appendix I: Additional Commissioning Checklists

Appendix I.1 - Electrical Level 3 System Test Checklist

- Switchgear protective relay settings verified against design coordination study.
- UPS module hot-swap tested with load bank; confirm no load interruption.
- Generator automatic start sequence tested from cold state; confirm time-to-rated-output within threshold.
- Automatic transfer switch tested for correct source prioritization and re-transfer logic.
- Ground fault protection tested at each distribution level.

Appendix I.2 - Mechanical Level 3 System Test Checklist

- Chiller staging sequence verified (lead/lag rotation, N+1 standby activation on fault).
- CRAH unit fan and valve modulation tested against BMS setpoint commands.
- Economizer changeover tested (or simulated if ambient conditions do not allow live test).
- Leak detection cable tested at multiple points per protected zone.
- Containment door interlocks and fire-mode release tested.

Appendix I.3 - Network Level 3/4 Test Checklist

- Every leaf-to-spine link verified for correct speed, duplex, and error-free operation.
- BGP underlay sessions verified established on all designed peerings.
- EVPN/VXLAN overlay tested with cross-rack Layer 2 adjacency validation.
- Single spine switch failure injected; confirm fabric reconvergence within design threshold.
- Border/edge router failover tested against both carrier circuits independently.

Appendix I.4 - Security Systems Test Checklist

- Every badge reader and mantrap tested for correct grant/deny behavior against test credentials.
- CCTV coverage walked and verified against design coverage map, no blind spots at entrances.
- Intrusion detection (door contacts, motion) tested at all monitored points.
- EPO and life-safety interlock tested per Section 9A.5.

Appendix J: Sample Rack Elevation Convention

Rack elevations in the DCIM platform follow a standard visual and data convention so that any engineer can interpret an elevation without site-specific training. U-position 1 is the bottom of the cabinet; devices are represented by their footprint in U-height with power feed (A/B), network connections, and asset tag visible in the device detail view.

U Position	Device Example	Power Feed	Notes
42-40	Top-of-Rack Leaf Switch (redundant pair)	A/B	Dual switches for NIC-level redundancy
39-30	Compute nodes (9x 1U)	A/B per node	Standard compute reference SKU
29-20	Compute nodes (9x 1U)	A/B per node	Standard compute reference SKU
19-8	Storage node (4U x 3)	A/B per node	Storage-dense reference SKU
7-1	Cable management / PDU	-	Vertical PDU mounted rear rail

Appendix K: Sample Operational Readiness Review Checklist

- All Level 1-5 commissioning activities closed with no open critical punch list items.
- All SOPs and MOPs for installed systems approved and version-controlled (Section 28.1).
- Full operations shift roster hired, trained, and certified (Section 27.3).
- DCIM/CMDB populated and verified accurate against physical as-built installation.
- Monitoring and alerting stack validated against all commissioning test scenarios (Section 29).
- Vendor support contracts and SLAs active and confirmed with vendor (Section 35).
- DR runbooks drafted and initial failover test scheduled (Section 32.2).
- Security systems and access control fully provisioned and access reviews scheduled (Section 17.4).
- Steering Committee go-live sign-off obtained.

Appendix L: Revision History

Version	Description	Status
0.1	Initial draft structure and Part I-II content	Superseded
0.5	Full draft, all parts, for engineering review	Superseded
1.0	Master reference document, current release	Current

Appendix M: Sample Vendor SLA Excerpt (Electrical Maintenance)

Clause	Term
Scope	Preventive and corrective maintenance for UPS, switchgear, and generator plant per Section 30.2
P1 Response Time	Remote response within 30 minutes, on-site within 4 hours (Section 35.2)
P1 Resolution Target	Restore full redundancy within 8 hours of arrival where parts are available on-site
Preventive Maintenance Cadence	Per Section 30.2 schedule; vendor to provide 2-week advance scheduling notice
Reporting	Post-visit service report within 2 business days, logged against DCIM asset record (Section 30.3)
Performance Review	Quarterly scorecard review per Section 35A.1

Appendix N: Escalation Contact Matrix Template

Role	Primary Contact Method	Escalation Trigger
On-call DC Operations Manager	Auto-page via ITSM (Section 29.2)	Any P1 incident
Facility Manager	Phone / auto-page	P1 incident exceeding 2 hours unresolved
Project Director / Steering Committee liaison	Phone	Facility-wide outage or safety incident
Security Lead	Auto-page via SIEM (Section 17.4.1)	Confirmed security incident
Primary electrical maintenance vendor	Vendor service line per SLA (Section 35.2)	P1/P2 electrical event
Primary mechanical maintenance vendor	Vendor service line per SLA (Section 35.2)	P1/P2 mechanical event

Appendix O: Extended Acronym Reference

Acronym	Meaning
A/B	Refers to the dual independent power or network paths (A path, B path)
BMS	Building Management System
CapEx / OpEx	Capital Expenditure / Operating Expenditure
CCTV	Closed-Circuit Television
CDN	Content Delivery Network
DCIM	Data Center Infrastructure Management
DMZ	Demilitarized Zone (network segment)
DR	Disaster Recovery
EPO	Emergency Power Off
EVM	Earned Value Management
FAT / SAT	Factory Acceptance Testing / Site Acceptance Testing
GPU	Graphics Processing Unit
IDS/IPS	Intrusion Detection System / Intrusion Prevention System
MTTR	Mean Time to Repair/Recovery
MV	Medium Voltage
NOC/SOC	Network Operations Center / Security Operations Center
PUE/WUE	Power Usage Effectiveness / Water Usage Effectiveness
QoS	Quality of Service
RACI	Responsible, Accountable, Consulted, Informed
RCM	Reliability-Centered Maintenance
SLA	Service Level Agreement
TCO	Total Cost of Ownership
WBS	Work Breakdown Structure

Appendix P: Document Cross-Reference Index (Selected Topics)

Topic	Primary Section	Related Sections
Tier III / redundancy philosophy	Section 10.1	6.1, 7.3, 20.3
Power Usage Effectiveness (PUE)	Section 7.4	1.1, 4.4, 36A.3
Network segmentation	Section 13.2	17.1, 17.2, 17A.1
Structured cabling standards	Section 14.1	Appendix H, Appendix J
Commissioning program	Section 20.3	Appendix C, Appendix I
Incident severity model	Section 29.2	31.1, 31A.1
Change management	Section 26.1	34.1, 17B.2.1
Disaster recovery	Section 32	32A, 12.4
Physical security layers	Section 9.1	9D, 9H
Capacity planning triggers	Section 20.4	33.2, 33A

Appendix Q: Assumptions Log

Assumption	Impact if Invalidated	Owner
Two diverse utility feeds available at selected site	Redesign of 2N electrical topology (Section 6.1)	Facility Design Lead
Three diverse carriers available at selected site	Redesign of WAN redundancy (Section 13.6)	IT/Network Architecture Lead
18-month schedule achievable given long-lead items	Re-baseline of Section 20 milestone schedule	PMO
Demand forecast in Section 3.1 holds within sensitivity band	Re-evaluation of phase trigger timing (Section 20.4)	Project Director

Appendix R: Sample Data Hall Environmental Log Template

Field	Description
Date/Time	Timestamp of reading (automated or manual round)
Data Hall / Row	Location identifier per Section 14.4.1 convention
Cold Aisle Temp	Measured against ASHRAE recommended range (Section 7.2.1)
Humidity	Measured against ASHRAE recommended range
Containment Status	Doors/panels intact, no breaches noted
Observations	Free text for any anomaly noted during rounds (Section 27A.2)
Recorded By	Shift technician identifier

Appendix S: Sample Change Request Form Fields

Field	Description
Change ID	Unique identifier, auto-assigned by ITSM platform
Requested By / Date	Requestor and submission date
System(s) Affected	Cross-referenced to CMDB asset records (Section 16, 34.2)
Change Type	Standard / Normal / Emergency (Section 34.1)
Risk Classification	Low / Medium / High (Section 28.3 framework applied to operational changes)
Implementation Plan	Reference to MOP or step-by-step plan
Rollback Plan	Required for Medium/High risk changes
Approval(s)	Per Section 26.1/34.1 thresholds
Scheduled Window	Date/time, cross-checked against change freeze calendar (Section 28B)
Post-Implementation Verification	Confirmation steps and sign-off

Appendix T: Sample Incident Report Form Fields

Field	Description
Incident ID	Unique identifier, auto-assigned by ITSM platform
Severity	P1-P4 per Section 29.2
Detection Time / Method	Automated alert vs. manual report
Systems Affected	Cross-referenced to CMDB (Section 16)
Incident Commander	Assigned per Section 31.1 (P1 only)
Timeline	Chronological log of detection, actions taken, resolution
Root Cause	Completed as part of post-incident review (Section 31.2)
Corrective Actions	Owner and target date, tracked in problem backlog (Section 31.3)

Appendix U: Reference Unit Conversion Table

Conversion	Factor
1 ton of cooling	≈ 3.517 kW
1 kW	≈ 3,412 BTU/hr
1 m ²	≈ 10.76 ft ²
1 kg	≈ 2.205 lb
1 L/kWh (WUE)	≈ 0.264 gal/kWh

Appendix V: Master Section Index

This document is organized into six parts and six primary sets of appendices. Part I (Sections 1-4, 2A-4D) covers project overview and business case. Part II (Sections 5-9, 9A-9Q) covers facility design and infrastructure. Part III (Sections 10-17, 17A-17S) covers IT and network infrastructure in the greatest depth, consistent with the primary engineering audience of this document. Part IV (Sections 18-26, 18A-26B) covers the project management plan. Part V (Sections 27-36, 27A-36A) covers operations and maintenance. Part VI (Appendices A-V) provides reference material, checklists, templates, and cross-reference indices supporting every preceding part.

How This Document Should Be Maintained

This document is a living reference: it is updated at each major design gate during delivery (Section 24.3), refreshed quarterly during operations to reflect actual as-built and as-operated conditions, and formally re-baselined at each phase trigger (Section 20.4) so that Parts II through V always reflect the current, accurate state of the facility rather than only its original design intent.

Appendix W: Full Job Description Summaries

Appendix W.1 - Facility Operations Technician

Field	Detail
Reports To	Facility Manager (Section 18.2, Part V Section 27.2)
Key Responsibilities	Rounds and inspections (27A.2), MOP execution (28), incident first response (31)
Required Qualifications	Electrical or mechanical trade certification, or equivalent experience
Required Training	Facility Operations Technician curriculum (Section 27B.1)
Shift Pattern	24x7 rotation per Section 27.2

Appendix W.2 - NOC/DC Operations Engineer

Field	Detail
Reports To	DC Operations Manager (Section 27.2)
Key Responsibilities	Monitoring and alert triage (29), incident/problem management (31), change execution support (34)
Required Qualifications	Network/systems engineering background
Required Training	NOC/DC Operations Engineer curriculum (Section 27B.2)
Shift Pattern	24x7 rotation per Section 27.2

Appendix W.3 - Security Officer

Field	Detail
Reports To	Security Lead (Section 18.2)
Key Responsibilities	Access control enforcement (9.3), CCTV monitoring (9.5), incident response (9D.3, 31)
Required Qualifications	Security licensing per local jurisdiction requirements
Shift Pattern	24x7 rotation per Section 27.2

Appendix W.4 - DC Operations Manager

Field	Detail
Reports To	Facility Manager / Operations Readiness Lead transition (Section 18.2, Part V)
Key Responsibilities	Team leadership, SLA/vendor management (35), incident commander for P1 events (31.1)

Field	Detail
Required Qualifications	Prior data center operations management experience

Appendix X: Sample Executive Summary Slide Content (for Steering Committee Kickoff)

- Project objective and success criteria (Section 1.1, 2.4)
- Business case headline: 10-year TCO and payback summary (Section 3.3.2, 3.3.3)
- Phased capacity plan: 1 MW to 4 MW over four phases (Section 1.1)
- 18-month schedule with key milestones (Section 20.2)
- Top 5 risks and mitigations (Section 23.2)
- Governance model and decision rights (Section 18)

Appendix Y: Full Standards Cross-Reference by Discipline

Discipline	Primary Standards Applied
Civil/Structural	Local building code, seismic code, IBC or regional equivalent
Electrical	NFPA 70 (NEC), IEEE 493, IEEE 1100, local utility interconnection standards
Mechanical	ASHRAE TC 9.9, local mechanical code
Fire/Life Safety	NFPA 75, NFPA 76, NFPA 2001, NFPA 13, NFPA 780, NFPA 70E
Structured Cabling	TIA-942, ANSI/TIA-568, BICSI 002
Security	CPTED principles, local security licensing regulation
Information Security	ISO/IEC 27001, SOC 2 (where applicable), Zero Trust reference architectures
Energy Management	ISO 50001
Business Continuity	ISO 22301
Availability Classification	Uptime Institute Tier Standard

Appendix Z: Full Table Index

This document contains detailed tables throughout Parts I-VI covering site scoring, financial models, redundancy architecture, equipment schedules, network and storage specifications, project schedules, budgets, risk registers, SOP/MOP indices, maintenance schedules, SLA references, and appendix reference material. Readers should use the Table of Contents and the cross-reference index in Appendix P to locate specific tables by topic rather than relying on this summary note alone.

Appendix AA: Document Approval

Role	Name (to be completed)	Signature / Date
Executive Sponsor		
Project Director		
Facility Design Lead		
IT/Network Architecture Lead		
Security Lead		
Operations Readiness Lead		

End of Document.

Appendix BB: Full Phase-by-Phase Capacity and Infrastructure Summary

Parameter	Phase 1	Phase 2	Phase 3	Phase 4
Critical IT Load	1.0 MW	2.0 MW	3.0 MW	4.0 MW
Data Halls Active	1	2	3	4
UPS Modules per Path	5 (N+1)	10 (N+1)	15 (N+1)	20 (N+1)
Generators (Duty + Standby)	3+1	4+1	5+1	6+1
Chillers (Duty + Standby)	2+1	3+1	4+1	5+1
Leaf Switches	24	48	72	96
Spine Switches	4	6	8	8 (400GbE upgrade)
Compute Racks (approx.)	20	42	65	88
Storage Racks (approx.)	6	12	18	24
Facility Ops Staff (per shift)	2	3	3	4
NOC Staff (per shift)	2	3	3	4

Appendix CC: Full Risk Register (Consolidated, All Categories)

ID	Risk	Category	Likelihood	Impact
R-01	Utility power delivery delay	Business Case	Medium	High
R-02	Construction cost escalation	Business Case	Medium	Medium
R-03	Migration schedule slippage	Business Case	Medium	Medium
R-04	Underutilization of built capacity	Business Case	Low	Medium
R-05	Regulatory/permitting delay	Business Case	Low	High
R-06	Geotechnical conditions differ from survey	Delivery	Low	Medium
R-07	Key trade contractor insolvency	Delivery	Low	High
R-08	Network/compute hardware chip shortage	Delivery	Medium	Medium
R-09	Commissioning reveals design/installation defect	Delivery	Medium	High
R-10	Operations staff hiring/training delay	Delivery	Low	Medium
R-11	Utility rate structure changes during delivery	Delivery	Low	Medium
R-12	Cybersecurity control gap found in independent audit	Delivery	Medium	High
R-13	Long-lead equipment delivery delay	Delivery	Medium	High
R-14	Labor/trade availability constraints	Delivery	Medium	Medium
R-15	Design changes during construction	Delivery	Medium	Medium
R-16	Utility outage exceeding fuel/runtime assumptions	Operations	Low	High

ID	Risk	Category	Likelihood	Impact
R-17	Vendor SLA non-performance	Operations	Medium	Medium
R-18	Staff turnover / knowledge loss	Operations	Medium	Medium
R-19	Capacity trigger missed / late procurement	Operations	Low	Medium
R-20	Supply chain compromise of hardware/firmware	Security	Low	High

Appendix DD: Full Glossary Continuation (Facility and Mechanical Terms)

Term	Definition
Air-Side Economizer	Cooling method using filtered outside air directly to cool the data hall when ambient conditions allow
Water-Side Economizer	Cooling method using cool ambient conditions to chill water without running compressors
Chilled Water Loop	Closed piping loop circulating chilled water between the chiller plant and air handling units
Condenser Water Loop	Loop rejecting heat from the chiller to cooling towers or dry coolers
Day Tank	Local fuel reservoir at a generator, refilled from bulk storage
Dry Cooler	Heat rejection device using ambient air without evaporative water use
Main-Tie-Main	Switchgear configuration with two mains and a tie breaker enabling either main to serve the full bus
Raised Access Floor	Elevated floor system creating an under-floor plenum, historically used for airflow and cabling
Rear-Door Heat Exchanger	Cooling unit mounted to the rear of a rack removing heat before it enters the room
Sound Attenuation Enclosure	Acoustic housing reducing generator noise to meet code/ordinance limits
Vibration Isolation	Mounting method reducing transmission of mechanical equipment vibration to the structure
Weathertight	Construction milestone indicating the building envelope excludes water intrusion

Appendix EE: Full Glossary Continuation (IT and Network Terms)

Term	Definition
Anti-Affinity Rule	Scheduling constraint preventing redundant workload instances from sharing a failure domain
Blast Radius	Scope of impact from a given failure or change
Border Leaf	Leaf switch role dedicated to external/WAN connectivity in a spine-leaf fabric
Canary Rollout	Deploying a change to a small subset of production before full rollout
CSI Driver	Container Storage Interface driver connecting a container platform to a storage backend
East-West Traffic	Network traffic between servers/racks within the data center, as opposed to north-south (in/out)
Erasur Coding	Data protection method distributing redundant data fragments across nodes/disks
Non-Blocking Fabric	Network design where any port can communicate with any other port at full rated speed simultaneously
Oversubscription Ratio	Ratio of downstream to upstream bandwidth capacity in a switch tier
Overlay Network	Virtual network built on top of a physical (underlay) network, e.g., VXLAN
Priority Class	Scheduling attribute determining preemption order under container platform resource contention
Underlay Network	The physical routed network beneath an overlay network

Appendix FF: Extended Standards Continuation

Standard	Scope
Uptime Institute Operational Sustainability	Staffing, maintenance, and operating procedure requirements complementing the Tier topology standard
ASHRAE 90.4	Energy standard for data centers (efficiency requirements)
IEC 62305	International lightning protection standard (alternative/complement to NFPA 780)
ISO/IEC 20000	IT service management standard, referenced for ITSM process alignment (Section 31, 34)
NIST Cybersecurity Framework	Reference framework for the security control categories in Section 17
PCI DSS	Payment card data security standard, applicable where in-scope workloads are hosted
HIPAA Security Rule	Applicable where healthcare data workloads are hosted (Section 3C.2)
GDPR / Regional Data Protection Law	Applicable where in-scope personal data workloads are hosted (Section 3C.2, 17R.2)

Appendix GG: Full Extended RACI (Governance-Level Activities)

Activity	Exec Sponsor	Project Director	PMO	Discipline Leads	Commissioning Agent
Charter approval	A	R	C	I	-
Site selection approval	A	R	C	C	-
Design freeze approval	I	A	C	R	C
Long-lead procurement approval	I	A	R	C	-
Budget baseline approval	A	R	R	C	-
Change request > \$250k	A	R	C	C	C (if Tier-impacting)
Level 4/5 commissioning acceptance	I	A	C	C	R
Go-live sign-off	A	R	C	C	C
Phase 2-4 trigger approval	A	R	R	C	-

Appendix HH: Sample Detailed IP Subnet Allocation (Phase 1, Illustrative)

Zone	Data Hall/Row	Subnet Size	Purpose
Production compute	DH01-R01 to R04	/23 per row group	Compute host addressing
Production storage	DH01 (all rows)	/24	SAN/NAS management + iSCSI/NVMe-oF
Container overlay	DH01 (cluster-wide)	/16 overlay CIDR	Pod/service networking (non-routed externally)
Out-of-band mgmt	DH01 (all rows)	/22	iDRAC/iLO, switch/PDU management
DMZ	Border/edge	/26 (public-facing)	Internet-facing services
Security/monitoring	DH01 (tap aggregation)	/25	SIEM, DCIM, NOC telemetry

Appendix II: Sample Commissioning Master Schedule (Level-by-Level)

Level	Scope	Typical Duration	Sign-off Authority
Level 1	Factory/component testing (off-site, per vendor FAT)	Ongoing through procurement window	Vendor + Project engineer witness
Level 2	Site acceptance testing of individual installed components	4-6 weeks	Discipline Lead
Level 3	System-level functional testing (e.g., full UPS system)	4-6 weeks	Discipline Lead + Commissioning Manager
Level 4	Integrated systems testing across electrical + mechanical	3 weeks	Commissioning Manager + Commissioning Agent
Level 5	Full integrated test under simulated failure scenarios	2-3 weeks	Commissioning Agent + Project Director

Appendix JJ: Closing Notes for Engineering Readers

This document consolidates design intent, delivery planning, and operating procedure into a single reference so that engineers joining the Project at any stage - design, construction, commissioning, or steady-state operations - can understand not only what a given system does, but why it was designed that way, how it was delivered, and how it is expected to be operated and maintained. Section-level cross-references throughout the document (see Appendix P) are intended to make it practical to jump directly to the relevant detail for a specific system rather than reading front-to-back for routine reference use.

Where a specific vendor model, exact contractual figure, or jurisdiction-specific code citation is required for execution, this document points to the authoritative source (vendor specification sheet, executed contract, or local code) rather than restating a figure that could drift out of date; the DCIM/CMDB (Section 16) and project document management system (Section 26B) remain the source of truth for as-built, as-operated detail over the life of the facility.

Appendix KK: Full Rack Elevation Reference - Standard Compute Row (Racks 01-06)

Rack	U42-U39	U38-U20	U19-U8	U7-U1
RK01	ToR Leaf A/B	Compute nodes (18x 1U)	Compute nodes (12x 1U)	PDU/Cable Mgmt
RK02	ToR Leaf A/B	Compute nodes (18x 1U)	Compute nodes (12x 1U)	PDU/Cable Mgmt
RK03	ToR Leaf A/B	Compute nodes (18x 1U)	Compute nodes (12x 1U)	PDU/Cable Mgmt
RK04	ToR Leaf A/B	Compute nodes (18x 1U)	Compute nodes (12x 1U)	PDU/Cable Mgmt
RK05	ToR Leaf A/B	Storage nodes (4U x 4)	Storage nodes (4U x 2)	PDU/Cable Mgmt
RK06	ToR Leaf A/B	Storage nodes (4U x 4)	Storage nodes (4U x 2)	PDU/Cable Mgmt

Appendix KK.1: Full Rack Elevation Reference - GPU/High-Density Row (Racks 21-24)

Rack	U42-U39	U38-U6	U5-U1
RK21	ToR Leaf A/B (100GbE)	GPU training nodes (6x 6U)	Liquid cooling manifold / PDU
RK22	ToR Leaf A/B (100GbE)	GPU training nodes (6x 6U)	Liquid cooling manifold / PDU
RK23	ToR Leaf A/B (100GbE)	GPU training nodes (6x 6U)	Liquid cooling manifold / PDU
RK24	ToR Leaf A/B (100GbE)	GPU training nodes (6x 6U)	Liquid cooling manifold / PDU

Appendix LL: Detailed Vendor Category Directory Template

Category	Primary Vendor (TBD)	Secondary/Backup Vendor (TBD)	Contract Reference
Utility power	-	N/A (single utility)	Interconnection agreement
Generators	-	-	Section 22.1 fixed-price contract
UPS systems	-	-	Section 22.1 fixed-price contract
Chillers	-	-	Section 22.1 fixed-price contract
Network hardware	-	-	Master purchase agreement
Compute hardware	-	-	Master purchase agreement
Storage hardware	-	-	Master purchase agreement
Structured cabling installer	-	-	Installation contract
Fire suppression	-	-	Installation + maintenance contract
Access control/CCTV	-	-	Installation + maintenance contract
DCIM software	-	-	Software license + support agreement
Managed SOC (if applicable)	-	-	Managed security services agreement

Appendix MM: Detailed Network Diagram Description - Spine-Leaf Fabric (Text Reference)

In lieu of a graphical network diagram, this appendix describes the logical topology in text form for reference alongside the as-built diagrams maintained in the DCIM platform (Section 16). Each of the four spine switches connects to every leaf switch in every data hall via a single 100GbE (Phase 1) uplink, giving every leaf switch four uplinks to four independent spines. Each rack is served by two leaf switches (A and B), each connected to all four spines, so a rack retains full fabric connectivity through the loss of any single spine switch or any single leaf-to-spine link. Border leaf switches connect the fabric to WAN/internet/cloud interconnects per Section 13.6, and are themselves deployed as a redundant pair following the same A/B path convention used throughout the electrical and network design (Section 10.1).

Appendix MM.1 - Logical Zone Diagram Description

The DMZ zone sits behind the border leaf and firewall pair, with a one-way, tightly filtered path into the production zone for the specific application flows requiring it. The production zone connects to the storage/replication network only through the storage fabric's own dedicated leaf switches, never sharing a broadcast domain with general user-facing VLANs. The out-of-band management zone is physically and logically separate, reachable only via jump host, and the security/monitoring zone receives mirrored/tapped traffic from every other zone without being able to originate traffic into them, consistent with Section 13.2.

Appendix NN: Sample Facility Floor Plan Description (Text Reference)

The technical building is organized with the four data halls (DH01-DH04) arranged along a central gray-space spine housing the electrical and mechanical plant rooms, allowing symmetric, minimal-distance power and chilled-water distribution to each data hall. The network meet-me room and carrier entrance facility sit adjacent to DH01 near the building's primary telecommunications entrance, with a diverse secondary entrance on the opposite side of the building serving the redundant carrier path described in Section 13.6.1. The security vestibule and visitor/vendor management area sit at the primary building entrance, with the loading dock and material handling path routed to avoid crossing through any data hall or security-sensitive corridor.

Appendix OO: Full Test Script Index for Level 5 Commissioning

Test Script ID	Scenario
L5-01	Total utility loss, both feeds, full facility generator transfer
L5-02	Single utility feed loss, confirm no impact on IT load
L5-03	Single generator fault during a live utility-loss event, confirm N+1 covers
L5-04	Single UPS module fault under full load, confirm no IT load impact
L5-05	Full A-path UPS system maintenance with full load on B path
L5-06	Single chiller fault at peak cooling load, confirm N+1 covers
L5-07	Single CRAH/in-row unit fault, confirm no thermal envelope excursion
L5-08	Combined electrical + mechanical fault (utility loss during chiller maintenance)
L5-09	Fire alarm activation, confirm suppression, EPO, and egress systems function as designed
L5-10	Single spine switch failure under production-representative load
L5-11	Full DCIM/monitoring platform failover test
L5-12	End-to-end DR failover test for a representative Tier 1 application

Appendix PP: Frequently Referenced Section Quick-Index

If you need to find...	Go to
Redundancy levels by system	Section 6.1.1
PUE/WUE targets and calculation	Section 4.4, 7.4
Network segmentation zones	Section 13.2
Alert severity definitions	Section 29.2
MOP risk classification	Section 28.3
Change approval thresholds	Section 26.1, 34.1
Backup/retention policy by tier	Section 12.4, 17W
Vendor SLA targets	Section 35.2, 35B
Commissioning levels explained	Section 20.3
Full glossary	Appendix A, G, DD, EE

Appendix QQ: Final Summary

This master reference document has traced the Data Center Project from its business justification through detailed facility and IT/network engineering design, through the plan for delivering it, and through the operating model that sustains it once live. Engineers using this document for detailed design work should treat Part III as the primary technical reference for compute, storage, network, and cabling decisions; engineers and technicians running the facility day to day should treat Part V and its associated SOP/MOP library as the primary operational reference; and project delivery staff should treat Part IV as the controlling reference for schedule, budget, and change governance through go-live.

As with any living engineering document, the authoritative state of the facility is always the as-built and as-operated record maintained in the DCIM/CMDB and document management systems described in Sections 16 and 26B, with this document serving as the structured, narrative companion to that operational record.

Appendix RR: Extended Preventive Maintenance Task Detail by System

Appendix RR.1 - UPS System Detailed PM Tasks

Task	Frequency	Estimated Duration
Visual inspection (connections, indicators, alarms)	Monthly	30 minutes per module
Infrared thermography of power connections	Quarterly	1 hour per UPS system
Battery/energy storage impedance testing	Annually	4 hours per string
Full load-bank test	Annually (rotating unit)	8 hours
Firmware/control software review	Semi-annually	2 hours
Capacitor and fan replacement (per vendor schedule)	Per vendor lifecycle guidance (typically 5-7 years)	1 day

Appendix RR.2 - Generator Detailed PM Tasks

Task	Frequency	Estimated Duration
No-load exercise run	Weekly	30 minutes
Fluid levels and visual inspection	Weekly	30 minutes
Fuel quality test/polishing	Quarterly	2 hours
Full load-bank test	Annually (rotating unit, N+1 maintained)	4-6 hours
Cooling system service	Annually	3 hours
Major overhaul (per vendor lifecycle guidance)	Per engine-hours threshold	Multi-day, scheduled well in advance

Appendix RR.3 - Chiller Plant Detailed PM Tasks

Task	Frequency	Estimated Duration
Visual inspection and log readings	Weekly	30 minutes per chiller
Refrigerant leak check	Quarterly	1 hour per chiller
Condenser/evaporator tube cleaning	Annually	1 day per chiller
Full mechanical service (compressor, controls)	Semi-annually	1 day per chiller
Vibration analysis	Annually	2 hours per chiller

Appendix SS: Extended Commissioning Defect Tracking Categories

Defect Category	Example	Typical Resolution Path
Installation workmanship	Incorrectly torqued electrical connection	Contractor rework before Level 2 acceptance
Design coordination	Piping routing conflicts with cable tray	RFI and design clarification (Section 22B.2)
Equipment defect	Factory defect found during SAT	Vendor warranty replacement
Control/sequence logic error	BMS staging sequence does not match design intent	Controls contractor reprogramming and retest
Documentation gap	As-built drawing does not match installed configuration	Contractor documentation correction before DCIM load

Appendix TT: Extended Security Control Mapping

Control Objective	Physical Control	Logical/Network Control
Prevent unauthorized facility access	Perimeter fencing, badge/biometric access (Section 9)	N/A
Prevent unauthorized data hall access	Mantrap, biometric verification (Section 9.4)	Access logged to SIEM (Section 17.4.1)
Prevent unauthorized network access	Out-of-band network isolation (Section 17A.1)	802.1X / NAC, firewall segmentation (Section 13.2, 17.1)
Prevent unauthorized administrative access	PAM-brokered access only (Section 17.2)	MFA, RBAC, just-in-time elevation (Section 17.4)
Detect anomalous behavior	CCTV analytics (Section 9.5)	SIEM correlation, UEBA where deployed (Section 17.4.1)
Protect data confidentiality	Restricted physical media handling (Section 17C.4)	Encryption at rest/in transit (Section 17.3)

Appendix UU: Extended Network Segmentation Firewall Rule Examples (Illustrative)

Rule ID	Source Zone	Destination Zone	Service	Action
FW-100	DMZ	Production - App Tier	HTTPS (443)	Allow (specific hosts only)
FW-101	Production - App Tier	Production - DB Tier	DB port (specific)	Allow (specific hosts only)
FW-102	Production - App Tier	DMZ	Any	Deny
FW-103	Non-production	Production	Any	Deny
FW-104	Out-of-band mgmt	Any (except jump host)	Any	Deny
FW-105	Any	Security/monitoring (tap)	Mirrored traffic	Allow (one-way)
FW-106	Internet	DMZ	HTTPS (443) only	Allow (rate-limited, WAF-inspected)

Appendix VV: Extended Sample Capacity Trigger Worksheet

Metric	Current	Threshold (Section 33.2)	Action Status
Critical IT load utilization	Tracked monthly	75% for 2 consecutive months	Reviewed at monthly capacity report (Section 33A.2)
Network port utilization (leaf)	Tracked monthly	75% for 2 consecutive months	Reviewed at monthly capacity report
Storage Tier 0 utilization	Tracked monthly	70% (below general 75%, given performance-tier sensitivity)	Reviewed at monthly capacity report
Rack space utilization (per data hall)	Tracked monthly	80%	Reviewed at monthly capacity report
Cooling headroom (per data hall)	Tracked continuously	80% of N capacity	Reviewed continuously via DCIM alarm (Section 16.2)

Appendix WW: Sample Onboarding Checklist for New Operations Staff

- Facility access provisioning (badge, biometric enrollment) per Section 17.4.
- Assignment of onboarding curriculum per role (Section 27B.1/27B.2).
- EHS orientation and PPE issuance (Section 27C.1).
- ITSM and DCIM platform account provisioning (Section 16, 29, 31).
- Shadow shift schedule assigned (minimum 4 weeks per Section 27B).
- Assignment of on-call rotation eligibility after certification (Section 27E.1).
- 30/60/90-day check-in scheduled with DC Operations Manager.

Appendix XX: Sample Vendor Onboarding Checklist

- Contract execution and insurance/bonding verification (Section 22B.1).
- Background check / security clearance for personnel requiring site access (Section 9D.2).
- Site-specific safety orientation (Section 27C.1).
- Access provisioning scoped to relevant racks/rows only (Section 9D.2, 17.4).
- SLA and escalation contact confirmation (Section 35.2, Appendix N).
- Addition to the vendor category directory (Appendix LL).

Appendix YY: Final Consolidated Reference Notes

Throughout this document, specific figures (capacity thresholds, staffing counts, cost percentages, schedule durations) are presented as representative, illustrative values consistent with an enterprise Tier III facility of the scale described in Section 1.1. Final figures for any specific number are confirmed during detailed design (Part II), detailed engineering within Part III, and the corresponding finance and procurement workstreams described in Part IV, and should be read as a structured starting point for that detailed work rather than as final, contractually binding values in themselves.

This structure - business case, facility design, IT/network design, delivery plan, and operating model, each cross-referenced against the others - is intended to remain the master organizing framework for the Project's documentation even as individual figures and specific vendor selections are finalized and updated throughout design, construction, and operations.